Supplementary file 2- Data extraction tables

Table 1 – Study source, year and systems aspects

	Study	Source	Year	Country	Systems -Approach**	Systems-Intervention***	Design	Risk
1	**Afsar-manesh et al.**[1]	Healthcare	2016	USA	1	1	0	0
2	**Allaudeen et al.**	Quality Management in Health Care	2017	USA	1	1	1	0
3	**Anderson et al.**	The Permanente Journal	2017	USA	1	1	1	0
4	**Bell et al.**	BMJ Tobacco Control	2017	UK	0	1	0	0
5	**Bhatt et al.**[3]	J. of Med. Syst.	2014	USA	1	1	0	1
6	**Bhutani et al.**[4]	J. of Obs., Gyn. & Neo. N	2006	USA	1	1	0	1
7	**Bowen et al.**	J. of Digital Imaging	2016	USA	1	0	1	0
8	**Bradley et al.**[5]	Int. J. of QHC	2011	Ethiopia	1	1	0	0
9	**Catchpole et al.**[6]	JAMA Surgery	2014	USA	1	1	1	0
10	**Chandrasekar et al.**	QJM: An Int. J. of Medicine	2017	UK	1	1	1	1
11	**Cochran et al.**	J. of Medical Systems	2018	USA	1	1	1	1
12	**DeFlitch et al.**	Health Env. Research & Design J.	2015	USA	1	1	1	1
13	**Dennerlein et al.**[7]	BMJ Occ. & Envir. Med.	2017	USA	1	1	1	0
14	**Gupta et al.**	J. of Oncology Practice	2018	USA	1	0	1	0
15	**Hathout et al.**[8]	Leadership in Health S.	2013	Canada	1	1	1	0
16	**Heymann et al.**[9]	Israeli Med. Ass. J.	2004	Israel	1	1	0	0
17	**Hultman et al.**	Annals of Plastic Surgery	2016	USA	0	1	1	0
18	**Huntington et al.**[10]	Bull. Of WHO	2012	Philippines	1	1	1	0
19	**Hwang et al.**[11]	E. J. of Emerg. Med.	2017	Korea	1	1	0	0
20	**Kane et al.**	Joint Com. J. Qual. & Patient Safety	2019	USA	1	1	1	0
21	**Khan et al.**	British Medical J.	2018	CANADA/USA	0	1	1	0
22	**Kottke et al.**[12]	The Permanente Journal	2016	USA	1	1	0	0
23	**Lick et al.**[13]	Critical Care Medicine	2011	USA	1	1	0	0
24	**Loh et al.**[14]	Int. J. of HCQA	2017	Singapore	1	1	1	1
25	**McGrath et al.**	IEEE J. of Biomedical & Health Infor.	2019	USA	1	1	1	1
26	**McKetta et al.**[15]	The Joint Co. JQ&PS	2016	USA	1	1	1	0
27	**Moran et al.**	E-Clinical Medicine	2018	UK	1	1	0	1
28	**New et al.**[16]	PLOS One	2016	UK	1	1	0	0
29	**Rateb et al.**[17]	Int. J. of HCQA	2011	Egypt	1	1	1	0
30	**Rothemich et al.**[18]	A. J. of Prev. Medicine	2010	USA	1	1	0	0
31	**Rustagi et al.**[19]	J. of AIDS	2016	Africa (3)*	1	1	1	0
32	**Ryan et al.**[20]	Drug & Alcohol Depend.	2006	UK	1	1	0	0
33	**Shultz et al.**[21]	A. J. of Public Health	2015	USA	1	1	1	0
34	**Srinivasan et al.**	Hospital Pediatrics	2017	USA	1	1	1	1
35	**Tetuan et al.**[22]	J. of Nursing Reg.	2017	USA	1	1	0	1

*Three African countries – Cote d'Ivoire, Kenya and Mozambique **Consideration of systems in the approach to developing the intervention ***Consideration of systems in the implementation of the intervention.

Supplementary file 2- Data extraction tables

Table 2 - Characteristics of studies [population and intervention]

Study	Clinical area	Participants	Health setting	Type of Systems Approach	How implemented	Length of time applied	Training to staff	Perception of Systems Approach	How developed
Afsar-manesh et al., 2016	Clinical Readmission rates	Staff (implementation) Patients (Data).	Whole Hospital-in General Medicine, General Surgery, Neurosurgery, Paediatrics, Orthopaedics	Comprehensive Lean methodology	System-wide leadership and promotion of improvement culture, patient-centeredness, process improvement and RCA in six clinical departments focused on reducing readmissions.	18 months	Created forum to share ideas and learn from colleagues and training in Lean principles.	System-wide with Lean principles	Used existing Lean principles
Allaudeen et al., 2017	Emergency care	ED patients and staff	University-affiliated department of veterans affairs medical centre	Lean-based multi-disciplinary initiative and PDSA	Delivered a rapid process improvement workshop to evaluate current processes, identified root causes of delays and developed counter-measures and standard work. The standard work was put into practice and monitored, feedback on success was obtained. Barriers to success were identified, and PDSA cycles were followed in response. Daily management systems to re-inforce, evaluate and refine standard work were also developed.	3 years 1yr pre-intervention period; 3yrs post-intervention period (this appears to include 1yr implementation period)	Not specified	Lean-based and multi-disciplinary	Standard approach – Lean
Anderson et al., 2017	Geriatric hip fracture	Geriatrics patients and staff including a clinical leadership team, clinical participants and senior management.	The University of Colorado hospital academic medical centre	"Step-wise framework for implementing a comprehensive geriatrics hip fracture program" involving twelve steps	A series of 12 steps, comprising elements such as: assembling a team, conducting a gap analysis, establishing reporting measures, designing and implementing interventions, and evaluating outcomes.	17 months Pre-intervention (1/1/2012 – 28/10/2014), Post intervention (29/10/2014 – 31/03/2016) Implementation period (03/2014 – 10/2014	Can't tell	Step-wise and comprehensive involving a wide range of stakeholders and taking account of local context.	Self-developed but seems to be informed by Kotter: Kotter JP. Leading change: Why transformation efforts fail. Harv Bus Rev 2007 Jan:1-10.
Bell et al., 2017	Antenatal care (smoking	Pregnant women	Eight acute NHS hospital trusts and 12	"BabyClear" – a complex intervention	Training for staff in participating agencies in skills, supporting	4 months	Training provided for staff in participating	Complex intervention, change in overall system of care - Multi-agency referral	Developed by Tobacco Control Collaborating Centre, part of a larger

Supplementary file 2- Data extraction tables

	cessation among pregnant women)		local authority areas in North East England	comprising a package of measures designed to support implementation of national guidance	materials, and implementation of referral pathway.	Implementation between 11/2012 and 07/2013 Pre and post-intervention	agencies in CO monitoring, communication skills, skills training	pathway with follow-up protocol	Improving Performance in Practice programme.
Bhatt et al., 2014	Operating Rooms	Nursing staff, ORs	Academic Medical Centre	ACGME Core Competency of Systems-based Practice	Process redesign involving problem definition, process changes and a multidisciplinary TT team and through horizontal Integration.	17 days	Intervention trailed. Surgical and TT team trained to implement the new system.	ACGME Systems-based Practice Team working and coordination.	Used existing ACGME Systems-Based Practice Pre intervention process analysed using structured approach.
Bhutani et al., 2006	Maternity & Neonatal – new-born Jaundice	Well babies discharged as healthy, Patients' parents, Paediatricians , Paediatric nurses, Home care nurse agencies, Lactation support services,	Semi-private urban birthing hospital	Systems approach to Clinical Condition Management	Incremental chronological adoption of each element: a) 1990-1992 b) 1993-1995 c) 1996-1998 d) 1999-2000 Assessment of entire process 2001-2003 Incremental implementation of a systems approach that incorporated a hospital policy to (a) authorize nurses to obtain a bilirubin (total serum/ transcutaneous) measurement for clinical jaundice, (b) universal pre-discharge total serum bilirubin (at routine metabolic screening), (c) targeted follow-up, using the bilirubin nomogram (hour-specific, percentile-based total serum bilirubin/ transcutaneous bilirubin), and (d) an organized institutional systems-based management of newborn jaundice	12 months	Parent education	An approach that relies on 1. Visual recognition 2. Measurement of bilirubin 3. Lactation and nutrition support 4. parent education including follow-up and is considered Systematic Multifactorial An 'approach that does not deteriorate over time and has institutional memory	Incremental changes to managing treatment of jaundice in new-borns Developed through literature review Systematisation of approach (algorithm generation)
Bowen et al., 2016	Stroke care	Stroke patients	Grady Memorial Hospital - Single centre, hospital, stroke centre	Multi-stakeholder process mapping to inform problem identification involving value stream mapping	Workflow process map was developed over a period of two months involving paging dispatcher, university call centre and emergency medical services manager. Included working with	32 days Pre-intervention (April 20 – May 6 2014),	Can't tell	Multiple stakeholder involvement in mapping processes to inform improvement - value stream mapping (Lean)	Standard method – process mapping

Supplementary file 2- Data extraction tables

					equipment vendor to test and confirm problem identified and supply appropriate equipment.	Intervention (May – Sept. 2014), Post-intervention (Sept. 17 – October 19 2014)			
Bradley et al., 2011	Primary care	Primary Healthcare Units (PHCU) – Patients	Primary care in Rural Ethiopia	The Ethiopian Millennium Rural Initiative "By systems-based, we mean healthcare improvement efforts that target all patients rather than those with specific diseases and that can be standardized and replicated across the country over time."	Through the elements of EMRI model: (i) improving the infrastructure of health centres (i.e. water, electricity, physical infrastructure and equipment), (ii) improvement in the supply chain (e.g. transport of specimens and results follow-up), (iii) human resource capacity building through health worker training and on-site clinical mentoring, (iv) developing a system to improve referrals between health posts and health centres and (v) community education and mobilization	18 months	Approach involved community education and mobilization Health worker training and on-site clinical mentoring.	A focus on health infrastructure, supply chain, human resource, between centre referral systems and community education and mobilization	Part of national health sector development efforts. No specific details As part of the Ethiopian Millennium Rural Initiative
Catchpole et al., 2014	Trauma care	Trauma Patients	Nonprofit, Academic tertiary care medical centre	Although the paper applies Human Factors Engineering, there is no clear emphasis on such an application as being part of systems engineering or systems approach. Yet, they unintentionally referred SEIPS and PDSA (iterative)	A multidisciplinary team was brought together for one and a half days to define problems and identify solutions. The main problem areas were identified, and a range of potential solutions to each were generated. Then, a short list was generated based on practical considerations or the projected time needed for implementation. This short list was framed within the components of the SEIPS model. After the meeting, members of the ED and trauma teams were invited to discuss the short list and be involved in the studies. As implementation moved forward, they used small, iterative PDSA to develop each intervention to a level	5 months	None	Unintentional. SEIPS just to frame potential solutions, to ensure coverage of task, team, environment and technology.	Used existing SEIPS human factors model

Supplementary file 2- Data extraction tables

					where it was practical and deliverable.				
Chandrasekar et al., 2017	Acute Kidney Injury	All medical inpatients at a single UK hospital	A university hospital in the UK	'QI Methodology' including driver diagrams, pareto charts and statistical process analysis Also includes a range of interventions involving risk assessment tools, early identification using automated alerts, development of an intervention bundle, formation of an outreach support team staff engagement and patient and family empowerment.	A multi-disciplinary project team was given remote QI training and met at a weekly huddle. Preparation strategies included mapping stakeholders, patient journeys and current processes. 3 years of data were analysed, a driver diagram and standard QI charter were developed to guide project. Also included: - staff education and awareness program, - development of a patient specific electronic alert to prompt diagnosis - implementation of a memorable AKI care bundle (ABCDE-IT), - creation of a new dedicated AKI outreach team - patient and family empowerment	34 months – Pre-intervention (Jan. 2011-oct. 2013), Post-intervention (Oct. 2013 – Jul. 2016) Entire project lasted 5 years	Remote training in QI for the team plus weekly huddles. Also staff engagement package, posters, seminars for key staff groups, formal and informal awareness events in the hospital	The project recognises multiple elements of the system as contributing to the problem (as evidenced by their driver diagram). The intervention attempted to address multiple elements within the system: staff knowledge, electronic patient health record prompts, new packages/care processes for patients with AKI Stakeholder mapping, mapping of patient journeys and identification of key care processes.	The project was triggered following a mortality analysis at the trust and joint collaboration with Institute of Healthcare Improvement in Boston, USA. The team went through remote training in QI, sought to influence business intelligence reporting and met at a weekly huddle.
Cochran et al., 2018	Emergency room	Patients, nurses and a team of ER and system engineering specialists	Franciscan Health Indianapolis – a general medical and surgical centre	Collective System Design (CSD) methodology involving PDCA – a systems engineering methodology, which recognises systems as the amalgamation of four key elements: Work/Actions, Structure, Thinking and Tone.	Senior leadership and ER team worked to identify the needs of internal and external customers, identified system boundaries, developed a CSD map and applied PDCA to design the relationships on the map for the purpose of implementation. Electronic logs of the medical centre was used to establish baseline.	8 months Pre-intervention (8 months), Post-intervention (8 months)	Can't tell	Recognition of a system as an amalgamation of four key elements that are always present and completely interrelated – work/actions, structure, thinking and tone or culture. Also defining stakeholders and the system boundary, understanding the needs of stakeholders, and determining the functional requirements to develop solutions.	Can't tell
DeFlitch et al., 2015	Emergency departmen t	Patients and staff	A suburban, tertiary care, academic ED, with paediatric and adult level 1 trauma.	"Engineering techniques" including defining a study team, process mapping, Discrete Event Simulation modelling and detailed design considerations	A study team was setup to carry out project. The team examined the operational data from our ED information system (EDIS), charted patient arrival patterns, conducted interviews of staff, observed staff with patients, and mapped the ED processes of care. Proposed model	3 years Pre-intervention (July 2005 – June 2006), Post-intervention	Student educational training	Engineering techniques that involve process mapping and simulation modelling and visualization of the operation of the system.	Use of existing tools - process mapping and simulation plus self-developed processes.

Supplementary file 2- Data extraction tables

				leading to the Physician directed Queuing (PDQ) model.	was tested in a simulation and piloted before implementation.	(July 2009 – June 2010) Intervention (July 2006 – June 2009)			
Dennerlein et al., 2017	Patient Handling and mobilisation	Direct Patient Care Workers	Hospital-wide -at 2 hospitals	A broad stakeholder engagement from senior level down, new lifting equipment across hospital, new processes and group training and one-to-one coaching and mentoring for staff	High-level buy-in with a multidisciplinary oversight committee chaired by the Associate Chief Nurse of Quality and a Collaborative Coordination Committee including Associate Chief, Occupational health ergonomists and nurse business officer The hospital expanded its investment in ceiling lifts, slings, sit-to stand devices and etc. The coordinating committee developed processes ensuring that all equipment was in working order and portable devices were stored on the units and readily available for use.	12 months	Yes, Programme training was provided to all nurses, nurse directors and patient care assistants. An external consultant provided an online introductory module, followed by group training and one-on-one coaching and mentoring at the bedside.	System-wide and multi stakeholder and multi processes Hospital-wide, and involvement of different stakeholders and multi components	Developed by a committee to include key component identified by previous systematic reviews, including an organisational policy aimed at reducing injuries, investment in equipment broad-based training within the context of providing tools and risk assessment
Gupta et al., 2018	Chemotherapy	Healthcare staff and patients	Parkland health and hospital system – a large public hospital in the USA	A multi-disciplinary team delivering PDSA including process mapping.	A multi-disciplinary team involving nurses, pharmacists, physicians, QI training programme coach, QI experts, IT analysts, unit secretary and patient representatives conducted assessment of existing waiting times, identified factors and conducted two cycles of PDSA.	6 months Pre-intervention (Jan. – Feb. 2017) Intervention (PDSA 1, Aug. – Sept. 2017; PDSA 2, Sept. – Oct. 2017) Post-intervention period not defined.	Can't tell	Focused on developing a preadmission process that streamlined patient evaluation on admission and improved communication.	Existing method – PDSA and process mapping.
Hathout et al., 2013	Sleep disorders	Healthcare staff	Province-wide, Manitoba, Canada	Stakeholder engagement, problem exploration, process mapping,	A project steering committee setup decided what is to be done. Consultations took place with stakeholders, staff, patients, administrators and managers.	18 months	Wide consultations, but PSG training was recommended as a result of the study	Stakeholder involvement, deep exploration of problem and system understanding (system	A multidisciplinary team was convened to improve the system to meet the population's needs.

Supplementary file 2- Data extraction tables

				exploration of systems drivers and value and objectives of services	A multi-disciplinary team was convened to improve the system to meet the population's needs. They articulated a vision, conducted a demand analysis, and then described the current state of the system. Using the demand analysis and their understanding of the current state they defined the desired state and worked through the process changes requirements to bridge the gap from the current state to the desired state.			drivers), its problems and stakeholders' needs.	
Heymann et al., 2004	Antibiotic overprescribing	Healthcare professionals-Staff	Maccabi Healthcare services, a Health Maintenance Organisation (HMO) serving 1.5M patients	Previously developed Systematic Inventive Thinking (SIT). Bases on "Creativity as an exact Science" by Genrich Altschuler	A multidisciplinary group was formed to work through the SIT steps – problem reformulation, general search strategy selection and application of idea-generation techniques. Results launched through national media campaign.	14 months	Not reported	Multiple stakeholder engagement, deep problem exploration, with focus on creative solutions. Unintentional, it has the elements of systems thinking, but the paper uses a systematic approach to solve complex problems	Based on previous work
Hultman et al., 2016	Breast reconstruction	Healthcare staff and patients	Academic medical centre of the University of North Carolina Hospitals	Lean-Six Sigma – using standard DMAIC model	A multi-disciplinary project team involving microsurgeons, anaesthesiologists, circulating nurses, surgical assistants liaising with other stakeholders.	24 months Pre-intervention (24 months), Intervention (10 months), Post-intervention (24 months)	Some team members were trained in six sigma with blue, green and yellow belts.	Six Sigma with multiple stakeholders	Existing method – Six sigma
Huntington et al., 2012	Maternal Health	Women's health teams	Health systems reform in a province in the Philippines involving two tertiary hospitals , 20 first level referral health facilities.	National initiative - National Safe Motherhood Programme. Seems influenced by WHO health systems strengthening principles	Implemented through national Department of Health initiatives Speed of implementation seems to have been the interventional factor. Fast in one province; normal in five provinces.	4 years	Intervention province reported that 74% of the referral providers had completed a competency-based clinical training programme. No information on clinical training was available for the comparison provinces.	Holistic understanding of a system's building blocks, identifying where a system succeeds, where it breaks down and what kinds of integrated approaches will strengthen the overall system.	No details reported. Appears influenced by WHO health systems strengthening principles.

Supplementary file 2- Data extraction tables

			Twelve rural health units. One barangay (neighbourhood or village) health station						
Hwang et al., 2017	Cardiac care	Patients	Multiple institutions in the chain of survival of cardiac arrest patients – community to hospital	System-wide CPR programme for OHCA patients developed by lead Hospital.	Started by identification of weak points in chain of survivor, CPR education sessions, improved records captured by EMS new protocol for ACLS at ED formulated by a multidisciplinary team.	12 months	CPR education for public at schools and workplaces.	"System-wide". Lots of emphasis on scope – who is involved – rather than how. Analysis of delivery system weaknesses in CA survival and multi interventions approach to address those weaknesses.	Developed in-house. No reference to any previous work.
Kane et al., 2019	Patient flow	Nursing bed managers, transfer line operators, patient pathway coordinators	A tertiary care hospital in Baltimore, USA	"A systems engineering approach" involving a steering group consisting of hospital CEO, CIO, COO, VP for medical affairs, Director of nursing, and project leaders. Involved working with external supplies and use of DES and ABS.	Governance structure and core leadership team set up for project, medical director playing a key leadership role. Core leadership group met weekly and drive strategic operational initiatives. There is also a dedicated data analytics team and a group of clinical representatives. Also involves partnership with GE Healthcare.	Not clear – it seems project started in 2014, various interventions implemented from 2014/15, 2016/17, 2018 and ongoing. Reported results based on preliminary data Entire project seems to have lasted 4 years	Can't tell	Specifically targeting patient flow throughout the hospital 'system'. Considers the context / wider system (e.g. outside treatment facilities). Considers key stakeholders, particularly in governance of the new system (see governance structure). Actively explores risks through two different types of complex simulation modelling.	Key stakeholders identified and then brought together to develop the command centre jointly. A new space was created so that key stakeholders could be physically collocated, SOPs created and then simulation modelling performed. Evaluation is still underway.
Khan et al., 2018	Paediatric inpatient unit	Patients, parents or caregivers, nurses, medical students and residents	Paediatric inpatient units in seven North American hospitals in USA and Canada	Co-production of intervention to standardise the structure of healthcare provider-family communication on family centred rounds.	A team of parents, nurses and physicians including health service researchers, medical educators, hospitalists, communication experts and health literacy experts coproduced the intervention to standardize healthcare provider-family communication on ward rounds ("family centered rounds"), which included structured, high reliability communication on bedside rounds emphasizing health	3 months - Pre-intervention (3 months), Intervention (9 months), Post - intervention (3 months)	A rounds training and learning programme for interprofessional team members.	Health service user centered (patients and families involved in production of the intervention). Intervention addressed multiple elements of the system i.e. targeted key stakeholders with education but also implemented process	Self-developed - A team of parents, nurses, and physicians, including health services researchers, medical educators, hospitalists, communication experts, and health literacy experts, coproduced the intervention—the Patient and Family Centered I-PASS

Supplementary file 2- Data extraction tables

					literacy, family engagement, and bidirectional communication.	Entire project lasted 25 months (December 2014 – January 2017)		changes, such as the mid-shift nurse-physician huddles. Intervention assessed across multiple domains, including reductions in medical errors and family experience.	
Kottke et al., 2016	Primary care - Coronary Heart Disease	Patients	Private, five-clinic primary care practice	Complex Adaptive Systems principles	Each clinic developed own system using systems' personnel including RN, Care Manager, IT staff and Clinic Assistant Care Coordinators. Activities related to patient care delivery, provider staff, staff education, training and tool development and information technology Through Team Based working.	6 months	Clinical service staff trained for use of previsit planning tool. Patient education materials. Including clinical based skills and CQI.	"1. health service delivery systems are complex adaptive systems, not mechanical systems, 2. Adoption of any system of care requires adaptation and reinvention and 3. The long-term survival of any system of care requires that a new process, at a minimum, does not threaten the viability of the overall system."	Using existing CAS theory to design interventions.
Lick et al., 2011	Cardiac Arrest	Patients	Community-based centres of excellence	"Take heart America programme". Community-based initiative involving 1. Widespread Cardiopulmonary resuscitation 2. Retraining of all emergency medical service personnel 3. Additional deployment of automated external defibrillators and 4. Protocol for transport to and treatment by cardia arrest centres.	Site coordinator appointed to work in lead hospitals in each of two counties. Coordinators established collaborations and implemented THA with city administrators, police and fire departments, school system administrators, survivors and survivor network organisations, ALS support team members, hospital administration and key clinicians in each Cardiac Arrest Centre (CAC).	6 months	Extensive Community CPR training, Public awareness, dispatcher instructed CPR. Advanced Life Support training for staff.	Emphasis appears to be on the wide coverage of the programme – community-side and multi-agency. Take Heart America (THA) model of improving care.	Developed by implementing all the high level 2005 American Heart Association (AHA) CPR and Emergency Cardiovascular Care Guidelines in a community-wide systems approach based on treatment models for other complex diseases such as HIV
Loh et al., 2017	Cataract Surgery	Patients	National tertiary specialist hospital	SEIPS Model/ PDSA	SEIPS used as framework for classification of problem, PDSA approach to improvement	6 months	Briefing of staff on data collection and weekly reminders	SEIPS framework	Existing approach Standard SEIPS model tailored to the case in question.

Supplementary file 2- Data extraction tables

					1. Retrospective study 2. Qualitative descriptions of incidents. 3. Applied SEIPS as a reference framework.				
McGrath et al., 2019	Postoperative surgical care& General medical inpatient care	Patients and staff including Nurses, Nurse Assistants, Occupational therapists, Physical therapists, Physicians	US Academic Health Centre	System-level design and analysis involving system design and validation, installation and education, operation and performance measurement	System design and validation phase was used to set goals for improvement based on organisational data and review of existing systems. Following elicitation of desired improvements and compilation of feature list, workshops were held with technical and clinical stakeholders to develop integration, installation, workflow and safety specifications and processes for selected system.	5 months - Pre-intervention (5 months), 2 months of intervention Post-intervention (5 months)	Education materials were created and delivered to staff to assist in understanding purpose, goals and to orient staff to new system, operation, workflows, and medical record processes.	Systematic technical and workflow design, implementation and performance measurement phases. Views the systems element as a preparation phase of exploration, piloting, and validation. Then moves into discrete implementation and measurement phases which seem separate to the systems element.	Not clear. No specific tools or approaches mentioned, and no clear grounding in systems literature.
McKetta et al., 2016	Paediatric Cardiac procedures	Physicians Nurses Technicians Improvement specialists	The Cardiac Centre at a Children's Hospital	A Discrete Event Simulation together with traditional QI involving a multidisciplinary team using a four-step framework – Define, Diagnose, Test and Implement, and Sustain. Including PDSA	Implementation led by a multidisciplinary team of physicians, nurse practitioners, nurses, technician and improvement support using an in-house framework- Define, Diagnose, Test and Implement, and Sustain. Tests were evaluated using PDSA cycles.	4 months	Not reported Daily debrief to sustain performance	Discrete Event Simulation (DES) as a tool for analysing complex systems Change management in complex systems. DES combined with QI a model for addressing this. Aim to maintain throughput during resource restriction (closed procedure suite).	A previously developed DES model was used and a four-step improvement framework developed in-house.
Moran et al., 2018	Major trauma	Population of England and wales- All hospitals in England and Wales (primary analysis done on 35 'constant submitter' units)	UK NHS in England and North Wales	Trauma systems – Systematic trauma care on a national basis.	NHS reorganisation creating a series of Regional Networks designated as Major Trauma Centres, with funding through a 'Best Practice Tariff' only available to MTCs over and above the normal funding for such patients. Data collected longitudinally through the Trauma Audit and Research Network (TARN).	4 years - Pre-intervention (Apr. 1st 2008 – Mar. 31st 2009) Intervention (2009/10 – 2011/12) Post-intervention (2013/14-2016/17)	Can't tell	Rationalised provision of trauma care through coordinated networks with an MTC hub.	Comparison with experience in the US. No clear reference to systems thinking literature.

Supplementary file 2- Data extraction tables

New et al., 2016	Trauma & Orthopaedics	Theatre staff	The Orthopaedic trauma theatre of a UK hospital Trust	A two-step intervention – one-day Lean training followed by 6 months coaching. Training covered Lean principles – Muda, Poka-Yoke, Genchi Genbutsu, Kaizen, flow, JIT, respect and teamwork, process mapping, PDCA cycles and a philosophy of continuous participative experimental improvement. Then a six-month improvement project.	The multidisciplinary team decided on improvement project after training and carried it out with support from experts.	6 months	A whole day training for a multidisciplinary staff team and practical training for project team during the improvement process Training in lean theory and methods with subsequent support and encouragement: one day training with light-touch coaching for six months (nurses, surgeons, anaesthetists and administrators)	A comprehensive Lean approach.	Existing method - Lean
Rateb et al., 2011	Health Insurance Organisation (HIO), pre-employment medical fitness check-up	Doctors, nurses, administrative staff, and customers	Egypt HIO / community, Medical fitness testing.	Business Process Re-engineering focusing on Structure, Process and Outcome. Systems approach appears to mean everything from building renovation to customer and staff satisfaction	Conducted brainstorming sessions involving stakeholders, decision makers, service providers and beneficiaries. Randomly selected six centres to take part in re-engineering phase which was implemented in three stages.	Can't Tell	New services, processes and standards introduced. IT training for staff	The entirety of Structure, Process and Outcome of care Business Process Re-engineering (BPR)	Approach developed by team using BPR concepts and Donabedian's model
Rothemich et al., 2010	US Family Practice / Public Health: smoking cessation	Adult smokers and Family physicians, general internists, nurse practitioners, physician assistants.	16 primary care practices	Called QuitLink Intervention. Limited details provided. Described as using paper-based, systems approach to identify smokers, provide advice to quit, and assess willingness to quit. Includes supporting	Selected practices were randomised into a control group and an intervention group. A nurse liaison provided training to all rooming staff at intervention practices on QuitLink implementation procedures. 1. Practice recruitment via researchers. 2. 2-month 'wash-in period' to incorporate methodology,	1 months	Training for staff, office managers and some clinicians Nurses and medical assistants trained in the QuitLink implementation: given a customised 'stamp', protocol process explained.	Ensuring communication from clinician to quitline and feedback from quitline to clinicians or a systematised population-health intervention with multiple points of action.	Self-developed: Synthesised from an evidence base around smoking cessation services – the approach used designed to address most of the perceived deficiencies in previous attempts.

Supplementary file 2- Data extraction tables

				willing smokers too access quitlines and communicate feedback from quitline to clinicians	install resources (quitline stamp and fax referral), train staff, obtain baseline data, and define analytic strata 3. Comparison period involving ongoing outcome measurement				
Rustagi et al., 2016	HIV AIDS	Healthcare staff and Patients	Mother-to-child HIV transmission prevention services in three countries in Africa – Cote d'Ivoire, Kenya and Mozambique	The Systems Analysis and Improvement Approach (SAIA) – a 5-step, iterative package of systems analysis and improvement tools developed using multiple systems engineering techniques including continuous quality improvement.	4-day workshops were held at each intervention facility to introduce and prepare staff for the intervention, follow-up visits were conducted weekly for 4 weeks, biweekly for 8 weeks after and then monthly visits thereafter or as needed by staff.	9 months	Training and support regular support provided for staff	Approach targets wider system from district level to local processes and action. Tools in SAIA include Cascade analysis tool – excel spreadsheet for quantitative analysis of patient flows, Value Stream Mapping (VSM) and PDSA	Self-developed based on multiple existing systems engineering tools.
Ryan et al., 2006	Alcohol detoxification	Service users	Manchester Alcohol Service (MAS) In-patient detoxification service	A whole systems approach to alcohol services – A collaborative working between multiple organisations	Implementation of approach occurred with new contracts issued to each of the providers: in-patient and home detoxification, community treatment, day care and access into rehabilitation services and other wrap-around services	Can't Tell	No	Collaborative working between organisations that individually addressed different parts of the needed service.	Previously developed and implemented MAS system. Current study only provides retrospective evaluation.
Shultz et al., 2015	Vaccine administration	Physicians and staff	5 Family Medicine Clinics and 4 Internal Medicine Clinics (as control)	Sequential and linked PDSA/Adjust cycles. A consensus-based framework that addresses the process of care.	Using collaborative working, five community-based family medicine clinics at the university of Michigan modified a point-of-care decision-support system for to improve administration and documentation Tetanus, diphtheria and acellular pertussis vaccines for patients.	Two years	Clinicians, nurses, medical assistants and support staff were trained to use the newly developed Automated Clinical Reminder (ACR) system.	A focus on Structure (physical environment and context of care), process (actions and procedures associated with the delivery and documentation of care) taking the needs of people into account.	An existing Automated Clinical Reminder (ACR) system was modified through consultations with clinicians, nurses, medical assistants, and support staff from each clinic.
Srinivasan et al., 2017	Paediatric inpatients	1 -23 month old Babies and parents and paediatric hospitalists,	Emergency department and inpatient unit of a 280-bed tertiary care, free-	Driver Diagram plus three cycles of PDSA involving stakeholder surveys focusing on changing clinician	A stakeholder survey was conducted and a multi-disciplinary team was set up. Stakeholder responses were turned into a driver diagram and projects for 3 PDSA cycles with a 3 week period at the	3 weeks - Pre-intervention (Jan. 2015 – Apr. 2015),	Face to face by study team to clinical providers during routine meetings, email communications to	Systems changes seem conceptually confined to process changes (NG feeding tube order set) and physical changes (stocking of ED with appropriate	Existing method – PDSA and Driver Diagram. Explicit reference to PDSA cycles but not to systems thinking/approaches

Supplementary file 2- Data extraction tables

		paediatric emergency medicine physicians, nurses, residents, interns, and nurse practitioners.	standing children's hospital	behaviour through both education, reinforcement and encouragement.	end of each cycle and a wide engagement of stakeholder with results of each cycle.	Intervention (Jan. 2016 – Apr. 2016) Post-intervention period unclear appears to be 3 weeks.	those not attending meetings, posters in clinical areas, pocket cards for clinicians, parental information sheets, walk-throughs 2-3 times per week to trouble shoot.	supplies). However, the driver diagram denotes different interventions having an impact on multiple possible drivers, more consistent with a systems thinking approach.	
Tetuan et al., 2017	Medication administration	Nurses	Integrated health care systems comprising primary and speciality clinics, and a 568-bed acute care hospital.	Systems Thinking Education Programme (STEP)	Medication huddles and monthly Organisation-wide education for 1yr. 1. Staff training (over 12 months) on systems thinking 2. Medication huddles 3. Observation audits of the medication administration process	12 months	Monthly training for 1yr Training of trainers for medication huddles, and direct subsequent training of other staff.	System-wide: Multifaceted intervention based around a definition of systems thinking as "the ability to recognise, understand, and synthesis the interactions and interdependencies in a set of components designed for a specific purpose".	Literature review of systems thinking, error detection, and safety culture.

Supplementary file 2- Data extraction tables

Table 3 – Characteristics of study cont. [Design, and other quality issues]

Study	Study Design	Baseline type (Prospective – study data OR Retrospective-routine data	Blinding of outcome measure (Yes, No, Can't Tell)	Funding source
Afsar-manesh et al., 2016	Before, During & After	Retrospective routine data	Can't Tell?	No external funding
Allaudeen et al., 2017	Before, During and After/ Concurrent Control	Retrospective routine data	Can't tell	Not reported
Anderson et al., 2017	Before, During and After	Retrospective routine data	Can't tell	Not reported
Bell et al., 2017	Before, During and After	Retrospective routine data	Can't tell	This article presents independent research funded by the NIHR School for Public Health Research (SPHR). NIHR SPHR is a partnership between the Universities of Sheffield, Bristol, Cambridge, Exeter, University College London; The London School for Hygiene and Tropical Medicine; the LiLaC collaboration between the Universities of Liverpool and Lancaster; and Fuse, the Centre for Translational Research in Public Health, a collaboration between Newcastle, Durham, Northumbria, Sunderland and Teesside Universities. Fuse is a UK Clinical Research Collaboration (UKCRC) Public Health Research Centres of Excellence, which receives funding from the British Heart Foundation, Cancer Research UK, Economic and Social Research Council, Medical Research Council, and the National Institute for Health Research.
Bhatt et al., 2014	Before, During & After	Retrospective routine data & Prospective study data	Can't Tell?	Lead author funded by hospital
Bhutani et al., 2006	Before, During & After	Prospective study data	Can't Tell?	Eglin fund and the New-born Paediatric Research Fund
Bowen et al., 2016	Before, During and After	Retrospective routine & Prospective Study data	Can't tell	Not reported
Bradley et al., 2011	Before, During & After	Retrospective routine data	Can't Tell?	The Children's Investment Foundation Fund
Catchpole et al., 2014	Before, During & After	Retrospective routine data & Prospective study data	No	Telemedicine and Advanced Technology Research Centre of the US Department of Defence
Chandrasekar et al., 2017	Before, During and After	Retrospective routine data	Can't tell	Not reported
Cochran et al., 2018	Before, During and After	Retrospective routine data	Can't tell	Not reported
DeFlitch et al., 2015	Before, During and After	Retrospective routine data	Can't tell	Received no financial support for the research authorship, and/or publication of this article.

Supplementary file 2- Data extraction tables

Dennerlein et al., 2017	Before, During & After	Retrospective routine data	Can't Tell?	National Occupational for Safety and Health for the Harvard T.H. Chan School of Public Health Centre for Work, Health and Wellbeing; National Institute of Arthritis and Musculoskeletal and Skin Disease of the National Institute of Health; Partners HealthCare
Gupta et al., 2018	Before, During and After	Retrospective routine data	Can't tell	Not reported
Hathout et al., 2013[8]	Before, During & After	Prospective study data	Can't Tell?	Not reported
Heymann et al., 2004[9]	Before, During & After	Retrospective routine data & Prospective study data	Can't Tell?	Not reported. Note – lead author works for HMO, programme evaluated by HMO
Hultman et al., 2016	Before, During and After	Retrospective routine data	Can't tell	Not reported
Huntington et al., 2012	Before, During & After /Concurrent Control	Retrospective routine data	Can't Tell?	World Bank; Manila Country Office; Department of Reproductive Health and Research, WHO, Geneva
Hwang et al., 2017	Before, During & After	Retrospective routine data	Can't Tell?	One author received grants from College of Medicine, Korea University and the Korea Centres for Disease Control and Prevention.
Kane et al., 2019	Before, During and After	Retrospective routine data*	Can't tell	Not reported
Khan et al., 2018	Before, During and After	Prospective Study data	Yes Yes	This project was supported by grant CDR-1306-03556 from the Patient-Centered Outcomes Research Institute (principal investigator: CPL). AK was supported by grant K12HS022986 from the Agency for Healthcare Research and Quality (principal investigator: Jonathan Finkelstein; Boston Children's Hospital, Boston, MA). JDB was supported by grant 5T32HS00063-21 from the Agency for Healthcare Research and Quality (principal investigator: Jonathan Finkelstein). The funders had no role in the design of the study; in the collection, analysis, or interpretation of data; in the writing of the report; or in the decision to submit the article for publication. Researchers were independent from funders and all authors had full access to the data and can take responsibility for the integrity of the data and the accuracy of the data analysis.
Kottke et al., 2016	Before, During & After	Retrospective routine data	Can't Tell?	National Heart and Lung Institute, National Institutes of Health, Bethesda, MD. Grant #R18HL096563 was the sole financial support for this project.
Lick et al., 2011	Before, During & After	Retrospective routine data and Prospective study data	No	Funding support from Medtronic Foundation, Medtronic Corporation, the CentraCare Health Foundation and the Unity and Mercy Hospital Foundations
Loh et al., 2017	Before, During & After	Retrospective routine data	No	Not reported
McGrath et al., 2019	Before, During and After/ Concurrent Control	Retrospective routine & Prospective Study data	Can't tell	Agency for Healthcare Research & Quality (AHRQ)
McKetta et al., 2016	Before , During & After	Retrospective routine data & Prospective study data	Can't Tell	Not reported
Moran et al., 2018	Before, During and After	Retrospective routine data	No	Performed independently without external funding

Supplementary file 2- Data extraction tables

New et al., 2016	Before, During & After /Concurrent Control	Prospective study data	Can't Tell	NIHR Programme Grant for Applied Research (RP-PG-0108-10020)
Rateb et al., 2011	Before, During & After	Retrospective routine & Study data	Can't Tell	Not reported
Rothemich et al., 2010	Concurrent Control (RCT)	Prospective study data	No	Funded by grant from the Agency for Healthcare Research and Quality (AHRQ) (5R21HS014854-02). One author owns stock in a Quitline service provider.
Rustagi et al., 2016	Before, During & After /Concurrent Control (RCT)	Retrospective routine & Study data	Can't Tell	Funded by Eunice Kennedy Shriver National Institute of Child Health and Human Development, the National Institute of Allergy and Infectious Diseases, the National Cancer Institute, the National Institute on Drug Abuse, the National Heart, Lung and Blood Institute and the National Institute on Aging of the US National Institutes of Health under award numbers R01HD075057 and P30AI027757 (awarded to the University of Washington Center for AIDS Research), as well as the Doris Duke Charitable Foundation's African Health Initiative (awarded to K.S. and M.F.C.), and the Fogarty International Center grant number K02TW009207 (awarded to K.S.).
Ryan et al., 2006	Before, During & After	Retrospective routine data	Can't Tell	Funded by grant from Turning Point
Shultz et al., 2015	Before, During & After/Concurrent Control	Retrospective routine data	Can't Tell	Not reported
Srinivasan et al., 2017	Before, During and After	Retrospective routine data	Can't tell	National Institutes of Health Clinical and Translational Science Award grant UL1 TR000448
Tetuan et al., 2017	Before, During & After	Prospective study data	Yes, for medication errors	Not reported

Supplementary file 2- Data extraction tables

Table 4a – Numbers and numerical values – Patient outcomes: Studies with Before, During & After Designs

Study	Outcomes measures	After	During	Before	P-values
Afsar-manesh et al., 2016	Overall 30-day readmission rate	11.3%, [n or N not reported]	-	12.1%, [n or N not reported]	*P*< 0.05
	Subgroups:				
	General Medicine	16.7%	-	17.9%	*P*< 0.05
	General Surgery	7.8%	-	9.9%	*P*< 0.05
	Neurosurgery	7.4%	-	9.6%	*P*< 0.05
	Paediatrics	9.8%	-	10.8%	*P*< 0.05
	Orthopaedics	6.8%	-	8.0%	*P*< 0.05
Allaudeen et al., 2017	-	-	-	-	-
Anderson et al., 2017	Laboratory evaluation for secondary causes of osteoporosis - n/N (%):				
	Completed blood cell count	116/117 (99%)	-	154/154 (100%)	*p=1.000*
	Basic metabolic panel with calcium	111/117 (95%)	-	151/154 (98%)	*p=0.963*
	Hepatic function panel	104/117 (89%)	-	74/154 (48%)	*p<0.001*
	25-hydroxyvitamin D	104/117 (89%)	-	105/154 (68%)	*p<0.001*
	Pharmacotherapy on discharge - n/N (%):				
	Calcium	116/117 (99%)	-	84/154 (55%)	*p<0.001*
	Vitamin D	112/117 (96%)	-	107/154 (70%)	*p<0.001*
	Antiosteoporosis	70/117 (85%)	-	34/154 (24%)	*p<0.001*
	Follow-up appointment completed within 30 days- n/N (%):				
	PCP (Internal to system)	13/117 (45%)	-	14/154 (26%)	*p=0.363*
	Metabolic Bone Clinic	32/117 (28%)	-	4/154 (3%)	*p<0.001*
	Orthopedics Clinic	96/117 (82%)	-	118/154 (77%)	*p=0.175*
Bell et al., 2017	Probability of quitting smoking by delivery date:				
	Adjusted Odds Ratio (95% CI)	1.81 (1.54 to 2.12)	-	0.13 (0.09 – 0.19)	*p<0.001*
Bhatt et al., 2014	-	-	-	-	-
Bhutani et al., 2006	Readmission rates for intensive phototherapy – n/N (%) [Rates estimated from graph. "Before" rate is for 1998. 1994-95 values available]	19/3,227 (0.59%)	27/3,168 (0.85%)	94/8,186 (1.15%)	-
	Extreme hyperbilirubinemia	0	0	0	-
Bowen et al., 2016	-	-	-	-	-
Bradley et al., 2011	-	-	-	-	-

Supplementary file 2- Data extraction tables

Catchpole et al., 2014	-	-	-	-	-
Chandrasekar et al., 2017	Reduction in in-hospital AKI mortality pre – post project 30-day mortality rate for AKI patients	23.2% 25.9% decrease	-	Data not given Data not given	*P<0.0001*
Cochran et al., 2018	Patient satisfaction with quality of care Percent of patients leaving without treatment	41st percentile [n or N not reported] 0.26%, [n or N not reported]	- -	20th percentile [n or N not reported] 1.50%, [n or N not reported]	*not reported* *not reported*
DeFlitch et al., 2015	Left without being seen Patient satisfaction	0.6%,[n or N not reported] 85th percentile [n or N not reported]	- -	5.7%, [n or N not reported] 17th percentile [n or N not reported]	*p<0.0001* *p<0.0001*
Dennerlein et al., 2017	-	-	-	-	-
Gupta et al., 2018	-	-	-	-	-
Hathout et al., 2013	Patients on recommended treatment	70%, n or N not reported	-	55%, n or N not reported	-
Heymann et al., 2004	-	-	-	-	-
Hultman et al., 2016	Partial or total flap loss – n/N (%) Take-back rates – n/N (%) Overall complication rates	*3*/46 (7%) *11*/46 (23.9%) *14*/46 (30%)	*3*/27 (11%) *6*/27 (20.7%) *9*/27 (33.3%)	*1*/39 (3%) *8*/39 (20.5%) *14*/39 (35.9%)	*Not significant* *Not significant* *Not significant*
Huntington et al., 2012	Number of maternal deaths/yr at intervention site– n/N (%) Maternal Mortality Rate (MMR)	18/*15,789* (0.114%) 114	24 [N not reported]	42/*16,535* (0.254%) 254	Not reported
Hwang et al., 2017	Good neurologic recovery at discharge (CPC 1, 2) –n/N (%) Number of patients admitted to ICU – n/N (%) Successful TH in cases of comma – n/N (%) Discharged from hospital alive – n/N (%)	24/282 (8.5%) 101/282 (35.8%) 33/96 (34.4%) 51/282 (18.1%)	5/117 (4.3%) 31/117 (26.6%) 2/31 (6.5%) 15/117 (12.8%)	6/182 (3.3%) 29/182 (15.9) 1/27 (3.7%) 16/182 (8.8%)	*p=0.001* *p*<0.001 *p*<0.001 <0.05
Kane et al., 2019	-	-	-	-	-
Khan et al., 2018	Family experience during rounds – top-box scores(95%CI): Understood what was said on rounds	N = 890 62.8% (53.7% - 71.1%)	-	N = 947 53.9% (44.6% - 63.0%)	*p=0.03*

Supplementary file 2- Data extraction tables

	Medical team explained well possible changes to child's condition	59% (?)	-	56% (?)	*not available*
	Satisfied with opportunity to ask questions on rounds	72% (?)	-	69% (?)	*not available*
	Medical team listened to family concerns	72% (?)	-	67% (?)	*not available*
	Family was included in decision making	59% (?)	-	56% (?)	*not available*
	Family felt important in their role on rounds	57% (?)	-	50% (?)	*not available*
	Family respectfully spoken to on rounds	79% (?)	-	78% (?)	*not available*
	Quality of communication during morning rounds	66% (?)	-	62% (?)	*not available*
	Family experience after rounds - top-box scores(95%CI):	N = 890		N = 947	
	Satisfaction with frequency of updates on child	54% (?)	-	49% (?)	*not available*
	Quality of update explanations	60% (?)	-	58% (?)	*not available*
	Inclusion in decision making later in day	55% (?)	-	53% (?)	*not available*
	Written communication - top-box scores(95%CI):	N = 890		N = 947	
	Frequency of written updates	33.7% (23.9% - 45.2%)	-	15.6% (8.9% - 25.9%)	*p<0.001*
	Understood written updates provided	57.9% (46.4% - 68.6%)	-	46.5% (34.2% - 59.3%)	*p=0.04*
	Communication with doctors - top-box scores(95%CI):	N = 890		N = 947	
	Shared understanding with doctors of medical plan	59.2% (49.9% - 67.8%)	-	54.0% (44.6% - 63.1%)	*p=0.14*
	Doctors addressed family concerns	65.9% (56.8% - 73.8%)	-	61.8% (52.5% - 70.3%)	*p=0.22*
	Doctors made family feel an important part of healthcare team	60.9% (49.2% - 71.4%)	-	57.7% (45.9% - 68.7%)	*p=0.34*
	Communication with nurses - top-box scores(95%CI):	N = 890		N = 947	
	Shared understanding with nurses of medical plan	65.4% (58.4% - 71.8%)	-	55.3% (48.0% - 62.4%)	*p=0.02*
	Nurses addressed family concerns	70.2% (62.9% - 76.6%)	-	61.2% (53.4% - 68.5%)	*p=0.02*
	Nurses made family feel an important part of healthcare team	70.7% (61.4% - 78.6%)	-	63.2% (53.5% - 71.9%)	*p=0.04*
	Teamwork amongst providers - top-box scores(95%CI):	N = 890		N = 947	
	Teamwork amongst doctors and nurses	62% (?)	-	59% (?)	*not available*
	Understanding - top-box scores(95%CI):	N = 890		N = 947	
	Understood overall reason for child's hospital stay	72% (?)	-	72% (?)	*not available*
	Understood what needed before child can return home from hospital	66% (?)	-	62% (?)	*not available*
	Overall quality - top-box scores(95%CI):	N = 890		N = 947	
	Overall quality of child's care	73% (?)	-	69% (?)	*not available*
	Quality of communication during hospital stay	59% (?)	-	55% (?)	*not available*

Supplementary file 2- Data extraction tables

	Overall medical errors: n/N (%) Rate/1000 patient days (95%CI) Non-harmful errors: – n/N (%) Rate/1000 patient days (95%CI) Harmful (preventable adverse events): – n/N (%) Rate/1000 patient days (95%CI) Non-preventable adverse events: n/N (%) Rate/1000 patient days (95%CI)	245/1532 (*16%*) 35.8 (26.9 – 47.7) 164/1532 (*10.7%*) 22.0 (15.1 – 32.1) 81/1532 (*5.3%*) 12.9 (8.9 – 18.6) 31/1532 (*2%*) 5.2(3.1 – 8.8)	- - - - - - - -	259/1574 (*16.5%*) 41.2(31.2 – 54.5) 139/1574 (*8.8%*) 20.0 (13.2 – 30.2) 120/1574 (*7.6%*) 20.7 (15.3 – 28.1) 72/1574 (*4.5%*) 12.6(8.9 – 17.9)	 *p=0.21* *p=0.50* *p=0.01* *p=0.003*
Kottke et al., 2016	Coronary Heart Disease: Composite Score (Rates of patients meeting composite goals for CHD (blood pressure <140/90 mmHg, low-density lipoprotein cholesterol level < 100 mg/dl, tobacco-free, and using aspirin unless contraindicated)) – n/N (%) Coronary Heart Disease: Aspirin compliance – n/N (%) Diabetes: Aspirin compliance – n/N (%) Diabetes: Composite score Proport to diabetic patients (meeting CHD goal plus haemoglobin A_{1c} concentration <8%) – n/N (%) Proportion of patients satisfied or very satisfied with preventive services received – n/N (%) Providers satisfied or very satisfied with preventive services – n/N (%)	*317*/529 (59.9%) *516*/529 (97.5%) *497*/509 (97.6%) *231*/509 (45.4%) *296*/320 (92.4%) *152*/205 (74.3%)	- - - - - -	*206*/511 (40.3%) *333*/511 (65.2%) *284*/485 (58.6%) *119*/485 (24.5%) *362*/455 (79.6%) *137*/231 (59.5%)	*<0.0001* *<0.0001* *<0.0001* *<0.0001* *Not significant* *p=0.0017*
Lick et al., 2011	Survival to hospital discharge of all patients after out-of-hospital cardiac arrest – n/N (%)	48/247 (19%)	-	9/106 (8.5%)	*p=0.011*
Loh et al., 2017	-	-	-	-	-
McGrath et al., 2019	-	-	-	-	-
McKetta et al., 2016	-	-	-	-	-

Supplementary file 2- Data extraction tables

Moran et al., 2018	Care processes in hospitals with <u>consistent submissions (patients with ISS ≥ 9)</u>:				
	Seen by consultant in ED, year-n/N (%)	16/17 – 10,943/19,197 (63%) 15/16 – 9,876/18,151 (61%) 14/15 – 8,963/16,414 (60%) 13/14 – 8,103/14,793 (60%)	12/13 – 6,169/11,708 (58%) 11/12 – 4,250/9679 (47%)	10/11–3,183/8626 (39.3%) 09/10–2,103/6957 (32%) 08/09–1,504/5338 (29%)	*Not reported*
	Intubated in ED, n (%)	2016/17 – 1,917 (11%) 2015/16 – 1,959 (12%) 2014/15 – 1,845 (12.4%) 2013/14 – 1,778 (13.2%)	2012/13 – 1,460 (13.7%) 2011/12 – 1,198 (13.3%)	2010/11 – 1,098 (13.6%) 2009/10 – 918 (13.6%) 2008/09 – 701 (13.6%)	*Not reported*
	Treated at MTC, n (%)	2016/17 – 14,247 (82%) 2015/16 – 13,279 (82%) 2014/15 – 11,873 (80%) 2013/14 – 10,790 (80%)	2012/13 – 8,212 (77%) 2011/12 – 6,750 (75%)	2010/11 – 6,113 (75%) 2009/10 – 5,058 (75%) 2008/09 – 3,757 (73%)	*Not reported*
	Blood given within 6h, n (%)	2016/17 – 423 (2.2%) 2015/16 – 470 (2.6%) 2014/15 – 405 (2.5%) 2013/14 – 391 (2.6%)	2012/13 – 372 (3.2%) 2011/12 – 259 (2.7%)	2010/11 – 283 (3.3%) 2009/10 – 270 (3.9%) 2008/09 – 118 (2.2%)	*Not reported*
	TXA if blood given, n (%)	2016/17 – 382 (90%) 2015/16 – 426 (91%) 2014/15 – 365 (90%) 2013/14 – 323 (83%)	2012/13 – 236 (63%) 2011/12 – 60 (23%)	2010/11 – 7 (2.5%) 2009/10 – 0 (0%) 2008/09 – 0 (0%)	*Not reported*
	Survival at discharge, n (%)	2016/17 – 17,451 (91%) 2015/16 – 16,424 (91%) 2014/15 – 14,878 (91%) 2013/14 – 13,388 (91%)	2012/13 – 10,568 (90%) 2011/12 – 8,808 (91%)	2010/11 – 7,895 (92%) 2009/10 – 6,313 (91%) 2008/09 – 4,891 (92%)	*Not reported*
	Time to death, median (IQR)	2016/17 – 8 (4-14) 2015/16 – 8 (4-14) 2014/15 – 8 (4-14) 2013/14 – 8 (4-13)	2012/13 – 7 (4-13) 2011/12 – 8 (4-13)	2010/11 – 8 (4-13) 2009/10 – 8 (4-14) 2008/09 – 8 (5-14)	*Not reported*
	Care process in <u>all hospitals (patients with ISS ≥ 9)</u>:				
	Intubated by Dr prehospital, n (%)	2016/17 – 44 (0.1%) 2015/16 – 73 (0.2%) 2014/15 – 99 (0.3%) 2013/14 – 80 (0.3%)	2012/13 – 73 (0.3%) 2011/12 – 41 (0.2%)	2010/11 – 80 (0.5%) 2009/10 – 80 (0.7%) 2008/09 – 50 (0.6%)	*Not reported*
	Seen by consultant in ED, n (%)	2016/17 – 18,797 (46.2%) 2015/16 – 17,691 (45.3%) 2014/15 – 16,111 (46.3%) 2013/14 – 14,406 (46.3%)	2012/13 – 11,531 (43.7) 2011/12 – 7,601 (34.6)	2010/11 – 5,217 (30.5%) 2009/10 – 3,218 (27.3%) 2008/09 – 2,188 (25%)	*Not reported*

Supplementary file 2- Data extraction tables

	Seen by consultant in ED if ISS >15, n (%)	2016/17 – 9,412 (56.8%) 2015/16 – 8,876 (56.4%) 2014/15 – 7,942 (57.8%) 2013/14 – 7,044 (57.7%)	2012/13 – 5,552 (54.8%) 2011/12 – 3,825 (43.7%)	2010/11 – 2,712 (38.2%) 2009/10 – 1,713 (34.6%) 2008/09 – 1,136 (31.9%)	*Not reported*
	Seen by consultant in ED if GCS <13, n (%)	2016/17 – 2,724 (76%) 2015/16 – 2,755 (74.8%) 2014/15 – 2,558 (74.6%) 2013/14 – 2,384 (75.4%)	2012/13 – 1,981 (72.9%) 2011/12 – 1,338 (62%)	2010/11 – 1,027 (58%) 2009/10 – 664 (52.2%) 2008/09 – 459 (47.4%)	*Not reported*
	Intubated in ED, n (%)	2016/17 – 2,929 (7.2%) 2015/16 – 2,976 (7.6%) 2014/15 – 2,850 (8.2%) 2013/14 – 2,700 (8.7%)	2012/13 – 2,386 (9%) 2011/12 – 1,898 (8.6%)	2010/11 – 1,639 (9.6%) 2009/10 – 1,248 (10.6%) 2008/09 – 951 (10.9%)	*Not reported*
	Survival at discharge, year-n/N (%)	16/17–40407/44059 (91.7%) 15/16–38733/42371 (91.4%) 14/15–34558/37725 (91.6%) 13/14–30808/33647 (91.6%)	12/13–25829/28239 (91.5%) 11/12–21385/23211 (92.1%)	10/11–16535/17956 (92.1%) 09/10–11129/12123 (91.8%) 08/09–8245/8903 (92.6%)	*Not reported*
	TXA given, year-n/N (%)	16/17–3,041/44069 (6.9%) 15/16–3,633/42371 (8.6%) 14/15–3,092/37725 (8.2%) 13/14–2,511/33647 (7.5%)	12/13–1217/28239 (4.3%) 11/12–304/23211 (1.3%)	10/11–24/17956 (0.1%) 09/10–1/12123 (0%) 08/09–2/8903 (0%)	*Not reported*
	Blood given within 6h, year-n/N (%)	16/17–672/44069 (1.5%) 15/16–810/42371 (1.9%) 14/15–714/37725 (1.9%) 13/14–633/33647 (1.9%)	12/13–639/28239 (2.3%) 11/12–396/23211 (1.7%)	10/11–374/17956 (2.1%) 09/10–333/12123 (2.7%) 08/09–174/8903 (2%)	*Not reported*
	TXA and blood given within 6h, year-n/N (%)	16/17–601/44069 (89.4%) 15/16–717/42371 (88.5%) 14/15–616/37725 (86.3%) 13/14–485/33647 (76.6%)	12/13–394/28239 (61.7%) 11/12–89/23211 (22.5%)	10/11–7/17956 (1.9%) 09/10–1/12123 (0.3%) 08/09–1/8903 (0.6%)	*Not reported*
	Time to death within 30 days, year-median (IQR), N	16/17–8 (4-14), 44069 15/16–8 (4-14), 42371 14/15–8 (4-14), 37725 13/14–8 (4-13), 33647	12/13–7 (4-13), 28239 11/12–8 (4-13), 23211	10/11–8 (4-13), 17956 09/10–8 (4-14), 12123 08/09–8 (5-14), 8903	*Not reported*
New et al., 2016	90-day readmissions – n/N (%)	102/*567* (18%)	-	94/*470* (20%)	*p=0.3000*
	Complications – n/N (%)	70/*583* (12%)	-	47/*470* (10%)	*p=0.070*
Rateb et al., 2011	Percentage of satisfied customers with:				
	Medical services - n/N (%)	*216*/251 (86%)	-	*22*/63 (34.3%)	*P*<0.001
	Housekeeping - n/N (%)	*225*/251 (89.7%)	-	*21*/63 (34%)	*P*<0.001

Supplementary file 2- Data extraction tables

	Staff communication - n/N (%) Accessibility - n/N (%)	*231*/251 (91.9%) *215*/251 (85.8%)	- -	*18*/63 (31.3%) *19*/63 (30%)	*P*<0.001 *P*<0.001
Rustagi et al., 2016	Proportion of HIV-positive pregnant women who received antiretroviral medications – n/N (%): Cote d'Ivoire Kenya Mozambique Mean proportion of HIV-exposed infants who received an HIV CPR screening test by 6 or 8 weeks of age: Cote d'Ivoire Kenya Mozambique	Overall - *13*/17 (77.7%) 100%, N = 5 73.4%, N = 6 63.5%, N = 6 Overall mean=46.1%, N=18 51.2%, N = 6 41.5%, N = 6 46.3%, N = 6	- - - - - - - -	Overall - *12*/18 (66.45%) 79.2%, N = 6 52.5%, N = 6 67.6%, N = 6 Overall mean=34.5%, N = 18 36.0%, N = 6 44.5%, N = 6 23.35%, N = 6	*p*=0.36 *p*=0.62 *p*=0.02 *p*=0.23 *p*=0.25 *p*=0.57 *p*=0.88 *p*=0.04
Ryan et al., 2006	Severity of Alcohol Dependence Questionnaire (SADQ) score – n/N (%): **Mild/Moderate:** Planned discharge Unplanned discharge **Severe:** Planned discharge Unplanned discharge Housing – n/N (%): **Stable:** Planned discharge Unplanned discharge **Unstable:** Planned discharge Unplanned discharge	 721/977 (73.8%) 256/977 (26.2%) 1965/*2,748* (71.5%) 785/*2,754* (28.5%) 2340/3,233 (72.4%) 893/3,233 (27.6%) 390/572 (68.2%) 182/572 (31.8%)	 - - - - - - - -	 124/191 (64.9%) 67/191 (35.1%) 102/*164* (62.0%) 65/*171* (38.0%) 787/1,168 (67.4%) 381/1,168 (32.6%) 243/395 (61.5%) 152/395 (38.5%)	 *P<0.012* *P<0.008* *P<0.001* *P<0.032*
Shultz et al., 2015	-	-	-	-	-
Srinivasan et al., 2017	Rate of NG hydration – n/N (%) **Primary outcome measure** NG tube placed for hydration – n/N (%) **Rate of NG complications** Aspiration- n/N (%) Death - n/N (%) Epistaxis - n/N (%) Displacement/removal - n/N (%)	53/91 (58%) 53 (58%) n = 53 0/53 (0%) 0/53 (0%) 0/53 (0%) 17/53 (32%)	53/91 (58%)	0/221 (0%) 0 (0%) 0/221 (0%) 0/221 (0%) 0/221 (0%) 0/221 (0%)	*Not reported* -

Supplementary file 2- Data extraction tables

Tetuan et al., 2017[22]	-	-	-	-	-

Supplementary file 2- Data extraction tables

Table 4b – Numbers and numerical values – Patient outcomes: Studies with Concurrent Control Designs

Study	Outcomes measures	Intervention	Control 1	Control 2	Control 3	P-values
Allaudeen et al., 2017	-	-	-	-	-	-
Huntington et al., 2012	Percentage of births delivered in health facilities per year Number of maternal deaths per year – n/N (%) [showing best three of five controls]	72%, n=7,017 18/*15,789* (0.114%)	46%, n=Not provided 9, n=Not provided	33%, n=Not provided 11, n=Not provided	28%, n=Not provided 16, n=Not provided	- -
McGrath et al., 2019	-	-	-	-	-	-
New et al., 2016	90-day readmissions – n/N (%) Complications – n/N (%)	102/*567* (18%) 70/*583* (12%)	55/*306* (18%) 32/*320* (10%)	- -	- -	*p=0.3000* *p=0.070*
Rothemich et al., 2010	-	-	-	-	-	-
Rustagi et al., 2016	Mean proportion of HIV-positive pregnant women who received antiretroviral medications: Cote d'Ivoire Kenya Mozambique Mean proportion of HIV-exposed infants who received an HIV CPR screening test by 6 or 8 weeks of age: Cote d'Ivoire Kenya Mozambique	Overall mean=77.7%, N = 17 100%, N = 5 73.4%, N = 6 63.5%, N = 6 Overall mean=46.1%, N = 17 51.2%, N = 5 41.5%, N = 6 46.3%, N = 6	Overall mean=65.9%, n=17 100%, n=5 38.5%, n=6 64.9%, n=6 Overall mean=32.0%, N = 17 42.6%, N = 6 19.2%, N = 5 32.1%, N = 6	- - - - - - - -	- - - - - - - -	- - - - - - - -
Shultz et al., 2015	Number of patients receiving Tdap vaccination-n/N (%): Follow-up Intervention year Base-line Number of patients receiving flu vaccination-n/N (%): Follow-up Intervention year Base-line	 6,978/14,748 (47.3%) 12,267/22,565 (54.4%) 3,976/25,584 (15.5%) 4 417/14,748 (30.0%) 9 301/22,565 (41.2%) 6 867/25,584 (26.8%)	 4 343/14,395 (30.2%) 3 806/17,043 (22.3%) 2 623/18,587 (14.1%) 6 743/14,395 (46.8%) 8 197/17,043 (48.1%) 6 738/18,587 (36.3%)	 - -	 - -	 *p<0.001* *p<0.001*

Supplementary file 2- Data extraction tables

Table 5a – Numbers and numerical values – Service, Resource or Cost outcomes: Studies with Before, During & After Designs

Study	Outcomes measures	After	During	Before	P-values
Afsar-manesh et al., 2016	-	-	-	-	-
Allaudeen et al., 2017	ED Length of Stay for medicine admissions, hrs- mean (SD) N Year 1 Year 2 Year 3	7.2 [n or N not reported] 7.9 7.1 6.4	-	8.7 [n or N not reported]	*P<0.001*
Anderson et al., 2017	Hospital LOS in days – mean, median (SD) N Time to surgery in hours – mean, median (SD) N 30-day all-cause readmissions – n/N (%) Disposition on discharge- n/N (%): Home with or without home healthcare Skilled nursing facility or nursing home Died Other Follow-up appointment scheduled before discharge – n/N (%): PCP (Internal or external to system) Metabolic Bone Clinic Orthopaedics Clinic	5.5, 5.0 (SD 2.22) N = 117 26.5, 22.3 (SD 17.5) N = 117 3/117 (2.7%) 19/117 (16%) 93/117 (79%) 0/117 (05) 5/117 (5%) 33/117 (28%) 62/117 (53%) 109/117 (93%)	- - - - - - - - - -	6.4, 5.0 (SD 4.87) N = 154 29.0, 22.5 (SD 24.9) N = 154 5/154 (3.2%) 33/154 (21%) 105/154 (68%) 1/154 (1%) 15/154 (10%) 23/154 (15%) 5/154 (3%) 126/154 (82%)	*p=0.004* *p=0.168* *p=0.520* *p=0.244* *p=0.244* *p=0.244* *p=0.244* *p=0.006* *p<0.001* *p=0.005*
Bell et al., 2017	Referral rates for cessation service overall – Incidence Rate Ratio (95% CI)	2.47 (2.16 – 2.81)	-	-	-
Bhatt et al., 2014	Operating Room (OR) Turnover Time (TT) Incidence of turnover time greater than or equal to 30 minutes – n/N (%)	mean=23m35s (SD=5m52s), n=17 (in Orthopaedic and vascular surgery only) *2*/17 (11.7%)	- -	mean=38m51s, (SD=14m39s), n=13 (Orthopaedic and vascular surgery only) *9*/13 (69.2%)	*p*< 0.001 *Not reported*
Bhutani et al., 2006	Use of hospital-based intensive phototherapy – n/N (%) Use of exchange transfusion Never events (TSB level greater than 30mg/dl)	156/11,995 (1.3%) 1(1 in 11,995 well babies) 0	159/6,395 (2.49%) 2(1 in 3,198 well babies) 0	446/8,186 (5.44%) 5(1 in 1,637 well babies) 0	- - -

Supplementary file 2- Data extraction tables

	Close calls (TSB level greater than or equal to 25mg/dl)	1 in 15,000	-	1 in 625	-
Bowen et al., 2016	Transmission error rates (of stroke alerts) via the pager system – n/N (%): ED NRR CR	 *4*/88 (5.1%) *17*/88 (18.8%) *1*/88 (1.1%)		 *30*/75 (40.0%) *17*/75 (22.7%) *9*/75 (12.0%)	 *p=0.0001* *p=0.004* *p=0.208*
Bradley et al., 2011	Antenatal care coverage – n/N (%)	140/140 (100%) β = 41.4, R^2 = 0.55	-	56/140 (40%)	*P<0.002*
	Skilled birth attendant coverage – n/N (%)	14/140 (10%) β = 2.6, R^2 = 0.50	-	7/140 (5%)	*p=0.015*
	Antenatal care HIV testing coverage	119/140 (85%) β = 26.1, R^2 = 0.54	-	70/140 (50%)	*p<0.001*
	Health post and health centre HIV testing coverage	β = 2.7, R^2 = 0.39	-	-	*p<0.001*
	Average outpatient visit at health centres	β = 0.4, R^2 = 0.65 N= 10 health centres	-	-	*p=0.276*
Catchpole et al., 2014	Number of flow disruptions in Computed tomography (CT): High level trauma	 mean=18.5 (SD=18.6, Range=1-50), median=9.00, n=13	-	 mean=25.6 (SD=32.4, Range=1-105), median=13.5, n=14	 -
	Low level trauma	mean=9.60 (SD=6.32, Range=1-27), median=8.00, n=107	-	mean=9.80 (SD=7.89, Range=1-65), median=8.00, n=72	-
	Time in the emergency department (ED) in minutes: High level trauma	 mean=123 (SD=76.1, Range=39-250), median=85, n=13	-	 mean=127 (SD=67.9, Range=38-291), median=119, n=14	 -
	Low level trauma	mean=80 (SD=52.5, Range=13-335), median=70, n=107	-	mean=96 (SD=55.9, Range=18-347), median=84, n=72	-
	Length of Stay (LoS) in days: Cohort with Major Risk of Mortality	 Kruskal-Wallis test: LoS = 69 (z=-2.49), n=508 median=5	-	 Kruskal-Wallis test: LoS = 74, n=510, median=8	 p=0.01
	Cohort with Extreme Risk of Mortality	LoS=25, median=8	-	LoS= 33, median=8	-

Supplementary file 2- Data extraction tables

Chandrasekar et al., 2017	Reduction in average length of stay for AKI pre – post project	2.6hrs (14.1%), [n or N not reported]	-	-	*p<0.0001*
Cochran et al., 2018	Median length of stay per patient in minutes	162, [n or N not reported]	-	202, [n or N not reported]	*Not reported*
	Median door-to-doctor time	13, [n or N not reported]	-	27, [n or N not reported]	*Not reported*
DeFlitch et al., 2015	Average waiting time in minutes	11 [n or N not reported]	-	66 [n or N not reported]	*p<0.0001*
	Door-to-doctor time- median (MAD*), N	20 min (15), N = 56,676	-	52 min (52), N = 44,720	*p<0.0001*
	Door-to-bed time- median (MAD), N	20 min (15), N = 56,676	-	225 min (172), N = 47,167	*p<0.0001*
	Total length of stay- median (MAD), N	3.7hr (2.9), N = 57,257	-	4.8hr (3.5), N = 46,775	*p<0.0001*
	Length of stay for - median (MAD), N:				
	Discharged	3.0 hr (2.2), N = 43,527	-	4.0 hr (2.6), N = 35,628	*p<0.0001*
	Hospitalized	7.1 hr (4.1), n = 10,353	-	9.2 hr (5.8), N = 9,109	*p<0.0001*
	Observed	11.2 hr (8.2), N = 2,565	-	20.5 hr (9.0), N = 1,715	*p<0.0001*
	Same day care	5.8 hr (3.4), N = 654	-	7.6 hr (5.7), N = 323	*p<0.0001*
	Annual ED visits	57,257 (% change = 22)	-	46,775	*p<0.0001*
	Number of ED beds	47 (% change = 21)	-	39	*p<0.0001*
	Ratio of visits to ED beds	1218 (% change = 2)	-	1199	*p<0.0001*
	Number of hospital beds	484 (% change = -3)	-	500	*p<0.0001*
	RN hr/day	398 (% change = 18)	-	328	*p<0.0001*
	*Median Absolute Deviation (MAD)				
Dennerlein et al., 2017	All injuries – n/N (%)	388/2131 (18.2%)	-	448/2149 (20.8%)	-
	Body part affected - Neck/Shoulder pain:				
	Count – n/N (%)	43/2131 (2.0%)	-	64/2149 (3%)	*p<0.05*
	Rate/100FTEs (95% CI)	2.0 (1.5 – 2.7)	-	3.0 (2.3 – 3.8)	
	Rate Ratio (95% CI)	0.678 (0.46 – 1.00)			
	Cause of injury - Lifting/exertion injuries				
	Count – n/N (%)	174/2131 (8.1%)	-	239/2149 (11.1%)	*p<0.05*
	Rate/100FTEs (95% CI)	8.2 (7.0 – 9.5)	-	11.1 (9.8 – 12.6)	
	Rate Ratio (95% CI)	0.73 (0.60 – 0.89)			
	Nature of injury - Pain and inflammation				
	Count – n/N (%)	119/2131 (5.6%)	-	153/2149 (7.1%)	*p<0.05*
	Rate/100FTEs (95% CI)	5.6 (4.7 – 6.7)	-	7.1 (6.1-8.3)	
	Rate Ratio (95% CI)	0.78 (0.62 – 1.00)			
Gupta et al., 2018	Median patient arrival time (time of day)	8:45AM, N = 28	-	12:43PM, N = 36	*Not reported*
	Overall median delay from admission to chemotherapy (hrs)	3.2hrs, N = 28	-	6.2hrs, N = 36	*Not reported*

Supplementary file 2- Data extraction tables

Hathout et al., 2013	Time from referral to sleep study Days to treatment starting after prescription generated Wait for treatment after study Annual studies	Median = 125 days, [n or N not reported] 21 days 21 days 4,289	- - - -	Median=600 days 90 days 180 days 1,347	- - - -
Heymann et al., 2004	Per-visit antibiotic purchasing for influenza visits	58.1 per 1000 visits	-	79.2 per 1000 visits	*p<0.0001*
Hultman et al. 2016	OR Time in minutes – mean (SD), N Perioperative Time in minutes - mean (SD), N Length of stay in days – mean (SD), N Physician revenue/minute hospital revenue/minute	606 (SD 146), N = 46 58 (17), N = 46 5.2 (2.3), N = 46 US $7.59 US $25.11	652 (SD 196), N = 27 65 (16), N = 27 5.6 (1.9), N = 27 - -	715 (SD 168), N = 39 73 (16), N = 39 6.3 (1.6), N = 39 US $6.28 US $21.84	*p<0.01* *p<0.01* *p<0.01* *p = 0.02* *Not significant*
Huntington et al., 2012	Percentage of births delivered in health facilities per year Volume of blood supplies received (as a proxy indicator for improvements in availability of essential medical products for maternal health services)	72%, [n or N not reported] 983 units	35%, [n or N not reported] 355 units	28%, [n or N not reported] 36 units	- -
Hwang et al., 2017	Proportion of Out-of-Hospital cardiac arrest patients receiving all CPR delivery enhancements Percentage of bystander CPRs without dispatcher assistance Proportion of no documented arrest rhythm by EMS Percentage of ACLS under capnography monitoring Percentage of extracorporeal CPR Percentage of successful therapeutic hypothermia in coma patients	*24*/282 (8.5%) *78*/282 (27.7%) *0*/282 (0.0%) *175*/282 (62.2%) *29*/282 (10.5%) *97*/282 (34.4%) *245*/282 (87.1%)	*3*/117 (2.6%) *32*/117 (27.4%) *4*/117 (3.4%) *64*/117 (55.1%) *9*/117 (7.7%) *8*/117 (6.5%) *67*/117 (57.1%)	*1*/182 (0.5%) *24*/182 (13.2%) *11*/182 (6.0%) *75*/182 (41.4%) 3/182 (1.4%) *7*/182 (3.7%) *112*/182 (61.5%)	*P<0.0001* *Not reported* *p=0.004* *p=0.008* *p=0.052* *p<0.001* *p=0.005*

Supplementary file 2- Data extraction tables

	Immediate coronary angiography for cases of presumed cardiac aetiology Number of patients who were admitted to the ICU	101/282 (35.8%)	31/117 (26.5%)	29/182 (15.9%)	*p<0.001*
Kane et al., 2019	Average hospital occupancy rate Average in-patient length of stay for department of medicine – days Time from request for admission bed till patient departs from emergency department at 92% occupancy - hrs	92% [n or N not reported] 5.3 [n or N not reported] 6.3 [n or N not reported]	- 5.5 [n or N not reported] -	85% [n or N not reported] 5.7 [n or N not reported] 9.7 [n or N not reported]	*Not reported* *Not reported* *Not reported*
Khan et al., 2018	-	-	-	-	-
Kottke et al., 2016	-	-	-	-	-
Lick et al., 2011	Interval from 911 to advanced life support at the scene in minutes – mean (SD) N Bystander cardiopulmonary resuscitation-n/N (%) Impedance threshold device use – n/N (%) In-hospital treatment of cardiac arrest patients who survived to hospital admission-n/N (%): In-hospital hypothermia Cardiac catheterization Implantable cardiac Defibrillator placed	7.2 (SD 3.6) N=247 *72*/247 (29%) *160*/247 (64.8%) *44*/95 (46%) *45*/95 (47%) *24*/95 (25%)	- - - - - -	7.5 (SD 3.5) N=106 *21*/106 (20%) *9*/106 (8.5%) 0/37 (0%) *8*/37 (22%) *5*/37 (14%)	*p=0.556* *p=0.86* - - *p<0.001* *p=0.17*
Loh et al., 2017	Number of intraocular (IOL) lens errors-n/N (%) Time between two IOL incidents Number of intraocular lens near miss error Intraocular lens implant error rates	10/39,390 (0.025%) 56 days 140/39,390 (0.36%) 2.54 per 10,000 cases N=39,390	1/7,475 (0.013%) 385 days 9/7,475 (0.12%) 1.32 per 10,000 cases, N=7,475	3/6,111 (0.049%) 35 days 36/6,111 (0.59%) 4.91 per 10,000 cases, N=6,111	- - - *Not stated*

Supplementary file 2- Data extraction tables

McGrath et al. 2019	Time required to obtain and record vital signs in seconds - mean	128.9 [n or N not reported] (t = 7.2416, df = 159.12)	-	178.8	*p<0.0001*
					Values for RRs
	Monitoring system utilization – mean (Std Err.), N:				
	Monitored hours/patient day	19.57 (0.18), N = 71	-	17.26 (0.58), N = 71	*p<0.0001*
	Monitored hours/month	19053.3 (308.9), N= 71	-	15931.25 (342.88), N = 71	*p<0.0001*
	Frequency of vital sign measurement – mean (Std. Err), N:				
	SpO2	6.7(0.026), N= 71	-	6.33(0.026), N = 71	*p<0.0001*
	Temperature	5.63(0.024), N= 71	-	5.81(0.025), N = 71	*p=1.000*
	Respiratory Rate	5.66(0.024), N= 71	-	6.15(0.026), N = 71	*p=0.0598*
	Pulse rate	7.49(0.028), N= 71	-	6.47(0.026), N = 71	*p=0.8820*
	Clinical Alarms - mean (Std. Err.), N				
	Clinical alarms/patient day	7.07 (0.46), N= 71	-	4.85 (1.11), N= 71	*p=0.0263*
	Short duration clinical alarms/patient day	5.5(0.3), N= 71	-	3.85(0.84), N= 71	*p=0.0695*
	Long duration clinical alarms /patient day	1.08(0.25), N= 71	-	0.79(0.23), N= 71	*p=0.0516*
	Duration of clinical alarms/patient day	93.79(9.78), N= 71	-	59.31(16.1), N= 71	*p=0.0002*
	Clinical alarms/monitored hour	0.4 (0.02), N= 71	-	0.32 (0.08), N= 71	*p=0.1090*
	Short duration clinical alarms/monitored hour	0.31(0.02), N= 71	-	0.25(0.06), N= 71	*p=0.2200*
	Long duration clinical alarms/monitored hour	0.06(0.01), N= 71	-	0.05(0.02), N= 71	*p=0.2467*
	Duration of clinical alarms/monitored hour	5.33(0.523), N= 71	-	3.89(1.1), N= 71	*p=0.0020*
	Non-clinical Alarm - mean (Std. Err.):				
	Nonclinical alarms/patient day	29.89 (2.4), N= 71	-	16.78 (2.11), N= 71	*p<0.0001*
	Short duration nonclinical alarms/patient day	22.63(1.81), N= 71	-	9.7(1.41), N= 71	*p<0.0001*
	Long duration nonclinical alarms/patient day	2.67(0.26), N= 71	-	1.45(0.26), N= 71	*p<0.0001*
	Duration of nonclinical alarms/patient day	1679.53 (185.69), N= 71	-	24357.56(1708.62), N= 71	*p<0.0001*
	Nonclinical alarms/monitored hour	6.39 (0.8), N= 71	-	3.7 (0.53), N= 71	*p<0.0001*
	Short duration nonclinical alarms/monitored hour	4.84(0.63), N= 71	-	2.14(0.34), N= 71	*p<0.0001*
	Long duration nonclinical alarms/monitored hour	0.57(0.14), N= 71	-	0.32(0.09), N= 71	*p<0.0001*
	Duration of nonclinical alarms/monitored hour	359.24(42.78), N= 71	-	5373.81(562.91), N= 71	*p<0.0001*
	Patient information present in monitoring system:				
	Last name (%)	678/678 (100%)	-	*551*/557 (98.92%)	*p=0.0083*
	First name (%)	678/678 (100%)	-	*188*/557 (33.75%)	*P<0.0001*
	Room and bed (%)	678/678 (100%)	-	*319*/557 (57.2%)	*P<0.0001*
McKetta et al., 2016	Mean turnaround time in Catheterization labs in minutes – mean (SD) N	32 (SD 12) N= 138	-	55 (SD 34) N=135	*p<0.001*
Moran et al., 2018	Care process in hospitals with consistent submissions (patients with ISS ≥ 9):				
	First hospital MTC, n (%)	2016/17 – 12,513 (72%) 2015/16 – 11,468 (70%)	2012/13 – 7,078 (66%) 2011/12 – 5,496 (61%)	2010/11 – 4,813 (59%) 2009/10 – 3,885 (58%)	*Not reported*

Supplementary file 2- Data extraction tables

		2014/15 – 10,217 (69%) 2013/14 – 9,322 (69%)		2008/09 – 2,736 (53%)	
	Time to arrival, hours, median (IQR)	2016/17 – 1.7 (1.2-2.6) 2015/16 – 1.7 (1.2-2.5) 2014/15 – 1.6 (1.1-2.3) 2013/14 – 1.5 (1.1-2.3)	2012/13 – 1.3 (0.9-2) 2011/12 – 1.2 (0.8-1.8)	2010/11 – 1.2 (0.8-1.7) 2009/10 – 1.2 (0.9-1.7) 2008/09 – 1.2 (0.8-1.7)	*Not reported*
	CT at any time, n (%)	2016/17 – 13,868 (72%) 2015/16 – 12,818 (71%) 2014/15 – 11,276 (69%) 2013/14 – 9748 (66%)	2012/13 – 7371 (63%) 2011/12 – 5954 (62%)	2010/11 – 4874 (57%) 2009/10 – 3766 (54%) 2008/09 – 2690 (50%)	*Not reported*
	Time to surgery, median (IQR)	2016/17 – 22 (10.9-49) 2015/16 – 22 (11-47) 2014/15 – 21 (10-48) 2013/14 – 21 (10-48)	2012/13 – 20 (7-45) 2011/12 – 19 (6-46)	2010/11 – 18 (6-45) 2009/10 – 18 (5-46) 2008/09 – 18 (5-50)	*Not reported*
	Admitted to ICU or HDU, n (%)	2016/17 – 4595 (24%) 2015/16 – 4638 (26%) 2014/15 – 4151 (25%) 2013/14 – 3696 (25%)	2012/13 – 3101 (27%) 2011/12 – 2982 (31%)	2010/11 – 2719 (32%) 2009/10 – 2288 (33%) 2008/09 – 1656 (31%)	*Not reported*
	LOS in hospital, median (IQR)	2016/17 – 9 (5-19) 2015/16 – 9 (5-19) 2014/15 – 9 (5-19) 2013/14 – 9 (5-18)	2012/13 – 9 (5-19) 2011/12 – 9 (5-18)	2010/11 – 10 (5-19) 2009/10 – 10 (5-21) 2008/09 – 10 (5-21)	*Not reported*
	LOS in ICU/HDU, median (IQR)	2016/17 – 3 (1-8) 2015/16 – 3 (1-8) 2014/15 – 3 (1-8) 2013/14 – 3 (1-8)	2012/13 – 3 (1-9) 2011/12 – 3 (1-8)	2010/11 – 4 (2-10) 2009/10 – 4 (2-10) 2008/09 – 4 (2-10)	*Not reported*
	Care process in all hospitals:				*Not*
	First hospital MTC, year- n/N (%)	16/17 – 16,871/41149 (41%) 15/16 – 15,694 (40%) 14/15 – 14,139 (40.6%) 13/14 – 12,588 (40%)	12/13 – 9694 (36.8%) 11/12 – 6876 (31%)	10/11 – 5572/17092 (32.6%) 09/10 – 4055 (34%) 08/09 – 2789 (32%)	*reported*
	Time to arrival, median (IQR)	2016/17 – 1.8 (1.3-2.8) 2015/16 – 1.7 (1.2-2.6) 2014/15 – 1.6 (1.2-2.4) 2013/14 – 1.5 (1.1-2.2)	2012/13 – 1.4 (1-2.1) 2011/12 – 1.2 (0.8-1.8)	2010/11 – 1.2 (0.8-1.8) 2009/10 – 1.2 (0.9-1.7) 2008/09 – 1.1 (0.8-1.6)	*Not reported*
	Arrival at first hospital midnight – 8.00am, n (%)	2016/17 – 7184 (16.3%) 2015/16 – 6845 (16.2%)	2012/13 – 4388 (15.5%) 2011/12 – 3641 (15.7%)	2010/11 – 2894 (16.1%) 2009/10 – 2049 (16.9%)	*Not reported*

Supplementary file 2- Data extraction tables

		2014/15 – 5972 (15.8%) 2013/14 – 5241 (15.6%)		2008/09 – 1556 (17.5%)	
	CT at any time, n (%)	2016/17 – 28,865 (65.5%) 2015/16 – 27,059 (63.9%) 2014/15 – 23,036 (61%) 2013/14 – 19,774 (58.8%)	2012/13 – 15,626 (55%) 2011/12 – 12,313 (53%)	2010/11 – 8984 (50%) 2009/10 – 5953 (49%) 2008/09 – 4035 (45%)	*Not reported*
	Admitted direct or transfer to MTC, n (%)	2016/17 – 19,811 (48.7%) 2015/16 – 18,747 (48%) 2014/15 – 16,837 (48.3%) 2013/14 – 15,076 (48.4%)	2012/13 – 11,803 (44.8%) 2011/12 – 8893 (40.5%)	2010/11 – 7383 (43.1%) 2009/10 – 5394 (45.7%) 2008/09 – 3879 (44.7%)	*Not reported*
	Time to surgery, median (IQR)	2016/17 – 23.3 (13.6-47.3) 2015/16 – 22.5 (13.2-45.4) 2014/15 – 22.1 (12.3-46) 2013/14 – 21.5 (11.1-45.8)	2012/13 – 20.4 (8.7-44) 2011/12 – 20.5 (8.2-45.4)	2010/11 – 19.35 (6.7-44.8) 2009/10 – 19.4 (6.4-47.2) 2008/09 – 19.9 (5.8-50.5)	*Not reported*
	Admitted to ICU or HDU, n (%)	2016/17 – 7582 (17.2%) 2015/16 – 7719 (18.2%) 2014/15 – 7024 (18.6%) 2013/14 – 6347 (18.9%)	2012/13 – 5559 (19.7%) 2011/12 – 5180 (22.3%)	2010/11 – 4266 (23.8%) 2009/10 – 3090 (25.5%) 2008/09 – 2219 (24.9%)	*Not reported*
	LOS in hospital, median (IQR)	2016/17 – 10 (5-19) 2015/16 – 10 (5-19) 2014/15 – 10 (5-19) 2013/14 – 9 (5-18)	2012/13 – 9 (5-18) 2011/12 – 9 (5-18)	2010/11 – 9 (5-19) 2009/10 – 10 (5-20) 2008/09 – 10 (5-21)	*Not reported*
	LOS in ICU/HDU, median (IQR)	2016/17 – 3 (1-7) 2015/16 – 3 (1-7) 2014/15 – 3 (1-7) 2013/14 – 3 (1-7)	2012/13 – 3 (1-8) 2011/12 – 3 (1-7)	2010/11 – 4 (2-9) 2009/10 – 4 (2-9) 2008/09 – 4 (2-9)	*Not reported*
New et al., 2016	NOTECHS Mean (SD) – a measure of operating teams' non-technical skills	77.84 (SD 11.59) N= 25	-	73 (SD 7.1) N= 17	*p=0.938*
	WHO Time Out attempt – Component of WHO surgical safety checklist – n/N (%)	24/25 (96%)	-	17/17 (100%)	*p=1.000*
	WHO Time-Out complete compliance – number of cases in which all three components of Time-Out were completed – n/N (%)	9/25 (36%)	-	3/17 (18%)	*p=0.621*
	WHO Sign Out – number observed	1/25 (4%)	-	0/17 (0%)	*p=1.000*
		6.59 (SD 3.95) N= 25	-	7.85 (SD 2.69) N= 17	*p=0.098*

Supplementary file 2- Data extraction tables

	Glitch rate/hour – these are deviations from recognised processes with potential to reduce quality or speed – mean (SD) N Length of stay in days (SD)	7.7 (SD 15) N= 292	-	10.3 (SD 25) N= 224	*p=0.396*
Rateb et al., 2011	Monthly customer flows-mean (SD) N	3,334.3 (SD 1,888.6) N= 6	-	1,747.3 (SD 1,932.4) N= 6	*p*<0.001
	Average customer compliance with booking system across six centres (%)	Mean=75.8%, N=6	-	Mean=52.1%, N=6	*p*<0.001
	Mean time spent per customer cycle in minutes (SD) N	18.3 (SD 5.5) N=212	-	48.8 (SD 14.5) N=63	*p*<0.001
	Appointment delays (days) - mean	6.2 days	-	18 days	-
	Percentage of satisfied staff with:				
	Crowdedness	*101*/101 (100%)	-	*15*/36 (40.7%)	*p*<0.001
	Process flow	*100*/101 (99.2%)	-	*15*/36 (42.6%)	*p*<0.001
	General satisfaction	*100*/101 (99.1%)	-	*16*/36 (44.4%)	*p*<0.001
	Administrative process	*100*/101 (98.6%)	-	*13*/36 (37%)	*p*<0.001
	Housekeeping	*99*/101 (98.5%)	-	*12*/36 (32.4%)	*p*<0.001
	Medical process	*99*/101 (98.4%)	-	*15*/36 (40.7%)	*p*<0.001
	Financial benefit	*94*/101 (93.0%)	-	*12*/36 (33%)	*p*<0.001
Rustagi et al., 2016	Mean proportion of pregnant women tested for HIV at antenatal care visit:	Overall mean=95.9%, N=18	-	Overall mean=90.5%, N=18	*p=0.97*
	Cote d'Ivoire	100%, N=6	-	94.2%, N=6	*p=0.25*
	Kenya	96.0%, N=6	-	86.8%, N=6	*p=0.30*
	Mozambique	91.7%, N=6	-	90.6%, N=6	*p=0.91*
Ryan et al., 2006	-	-	-	-	-
Shultz et al., 2015	Number of visits per patient for Tdap and flu vaccinations- mean (SD) N:	2.6 (SD 2.4) N = 39,882	2.9 (SD 2.7) N = 39,822	3.0 (SD 2.9) N = 39,882	*Not stated*
	Number of patients receiving Tdap vaccination – n/N (%)	*26,419*/67,914 (38.9%)	*27,573*/67,914 (40.6%)	*10,119*/67,914 (14.9%)	*Not stated*
	Number of patients receiving flu vaccination.	*26,011*/67,914 (38.3%)	*30,018*/67,914 (44.2%)	*20,918*/67,914 (30.8%)	-
Srinivasan et al., 2017	Have you ever used or placed an NG tube for hydration in an infant with bronchiolitis – n/N (%)				
	Physicians – YES	90/115 (78%)	-	23/114 (20%)	*p<0.001*
	Physicians – NO	25/115 (22%)	-	91/114 (80%)	*p<0.001*

Supplementary file 2- Data extraction tables

	Nurses – YES	62/97 (64%)	-	11/86 (13%)	*p<0.001*
	Nurses - NO	35/97 (36%)	-	75/86 (87%)	*p<0.001*
	What is your preference for hydration of an infant with bronchiolitis?				
	Physicians – n/N (%):				
	IV	6/115 (5%)	-	49/114 (43%)	*p<0.001*
	NG	49/115 (43%)	-	14/114 (12%)	*p<0.001*
	Either	60/115 (52%)	-	42/114 (37%)	*p<0.001*
	Nurses – n/N (%):				
	IV	28/97 (29%)	-	43/86 (50%)	*p=0.003*
	NG	9/97 (9%)	-	10/86 (12%)	*p=0.003*
	Either	59/97 (61%)	-	30/86 (35%)	*p=0.003*
	What are your concerns about using an NG tube for hydration?				
	Physicians – n/N (%):				
	NG tube may obstruct the nasal passage	24/115 (21%)	-	47/114 (41%)	*p=0.001*
	Risk of aspiration	21/115 (18%)	-	17/114 (15%)	*p=0.59*
	Accidental placement of the NG in the airway	16/115 (14%)	-	28/114 (25%)	*p=0.05*
	Parental resistance to the NG tube	73/115 (63%)	-	79/114 (69%)	*p=0.4*
	Other	23/115 (20%)	-	21/114 (18%)	*p=0.87*
	Nurses – n/N (%):				
	NG tube may obstruct the nasal passage	38/97 (39%)	-	49/86 (57%)	*p=0.02*
	Risk of aspiration	27/97 (28%)	-	36/86 (42%)	*p=0.06*
	Accidental placement of the NG in the airway	19/97 (20%)	-	8/86 (9%)	*p=0.06*
	Parental resistance to the NG tube	41/97 (51%)	-	44/86 (51%)	*p=0.24*
	Other	32/97 (33%)	-	13/86 (15%)	*p=0.006*
	Is NG an option for hydration in our hospital bronchiolitis guideline?				
	Physicians – n/N (%):				
	Yes	91/115 (79%)	-	29/114 (25%)	*p<0.001*
	No	0/115 (0%)	-	7/114 (6%)	*p<0.001*
	Can't tell	22/115 (19%)	-	77/114 (68%)	*p<0.001*
	Nurses – n/N (%):				
	Yes	71/97 (73%)	-	24/86 (28%)	*p<0.001*
	No	0/97 (0%)	-	3/86 (3%)	*p<0.001*
	Can't tell	25/97 (26%)	-	59/86 (69%)	*p<0.001*
	Is NG an option for hydration in the AAP bronchiolitis guideline?				

Supplementary file 2- Data extraction tables

	Physicians – n/N (%): Yes No Can't tell	 70/115 (61%) 2/115 (2%) 42/115 (37%)	 - - -	 48/114 (42%) 0/114 (0%) 65/114 (57%)	 *p=0.002* *p=0.002* *P=0.002*
Tetuan et al., 2017	Nurse workarounds – n/N (%)	175/1,998 (8.8%)	-	305/1,652 (18.5%)	*P*< 0.0001
	Safety Attitudes Questionnaire (SAQ) Score –mean (SD) N	4.05 (SD 0.547) N = 334	-	3.95 (SD 0.605) N = 585	*P*= 0.029
	Systems Thinking Scale (STS) score-mean (SD) N	64.90 (SD 8.5) N =334	-	63.39 (SD 9.36) N = 585	*P*= 0.013
	Medication events – n/N (%)	84/1,998 (4.2%)	-	156/1,652 (9.4%)	*p<0.001*
	Workaround with time, dose or omission error – n/N (%)	11/1,998 (6.3%)	-	13/1,652 (4.3%)	*p=0.3276*

Supplementary file 2- Data extraction tables

Table 5b – Numbers and numerical values – Service, Resource or Cost outcomes: Studies with Concurrent Control Designs

Study	Outcomes measures	Intervention	Control 1	Control 2	Control 3	P-values
Allaudeen et al., 2017	Reduction in ED length of stay for combined medicine and surgical admissions	0.7hrs (*p-0.003*), [n or N not reported]	0.0hrs (*p=0.2*), [n or N not reported]	-	-	*p=0.001*
Dennerlein et al., 2017	All injuries – n/N (%)	388/2131 (18.2%)	180/2414 (7.46%)	-	-	-
	Body part affected - Neck/Shoulder pain:					
	Count – n/N (%)	43/2131 (2.0%)	11/2414 (0.46%)	-	-	-
	Rate/100FTEs (95% CI)	2.0 (1.5 – 2.7)	0.46 (0.25 – 0.85)	-	-	-
	Cause of injury - Lifting/exertion injuries					
	Count – n/N (%)	174/2131 (8.1%)	48/2414 (1.99%)	-	-	-
	Rate/100FTEs (95% CI)	8.2 (7.0 – 9.5)	1.99 (1.50 – 2.64%)	-	-	-
	Nature of injury - Pain and inflammation					
	Count – n/N (%)	119/2131 (5.6%)	29/2414 (1.20%)	-	-	-
	Rate/100FTEs (95% CI)	5.6 (4.7 – 6.7)	1.20 (0.83 – 1.73)	-	-	-
Huntington et al., 2012	Volume of blood supplies received (as a proxy indicator for improvements in availability of essential medical projects for maternal health services)	983 units	941 units in Camarines Sur	-	-	-
	Number of women's health teams formed	871 teams	391 teams	[Other control provinces reported no data]	-	-
	Proportion of first level referral providers who completed a clinical training programme	74%, [n or N not reported]	[No data provided by control provinces]	-	-	-
	Facility-based delivery rate by province [showing best three of five controls]	72%, [n or N not reported]	46%, [n or N not reported]	34%, [n or N not reported]	33%, [n or N not reported]	-
McGrath et al., 2019	Frequency of vital sign measurement - mean (Std Err.), N:					
	SpO2	6.7(0.026), N = 71	6.24(0.027), N = 61	-	-	-
	Temperature	5.63(0.024), N = 71	5.57(0.026), N = 61	-	-	-
	Respiratory Rate	5.66(0.024), N = 71	5.83(0.026), N = 61	-	-	-
	Pulse rate	7.49(0.028), N = 71	7.06(0.029), N = 61	-	-	-
	System utilisation - mean (Std Err.), N:					
	Monitored hours/patient day	19.57(0.18), N = 71	12.98(0.58), N = 61	-	-	-

Supplementary file 2- Data extraction tables

	Monitored hours/month	19053.3(308.9), N = 71	5225.05(208.95), N = 61	-	-	-
	Clinical Alarms - mean (Std Err.), N: Clinical alarms/patient day	7.07(0.46), N = 71	5.73(0.63), N = 61	-	-	-
	Short duration clinical alarms/patient day	5.5(0.3), N = 71	4.52(0.49), N = 61	-	-	-
	Long duration clinical alarms /patient day	1.08(0.25), N = 71	1.06(0.09), N = 61	-	-	-
	Duration of clinical alarms/patient day	93.79(9.78), N = 71	73.84(9.7), N = 61	-	-	-
	Clinical alarms/monitored hour	0.4(0.02), N = 71	0.5(0.07), N = 61	-	-	-
	Short duration clinical alarms/monitored hour	0.31(0.02), N = 71	04(0.06), N = 61	-	-	-
	Long duration clinical alarms/monitored hour	0.06(0.01), N = 71	0.09(0.01), N = 61	-	-	-
	Duration of clinical alarms/monitored hour	5.33(0.52), N = 71	6.47(1.22), N = 61	-	-	-
	Non-clinical Alarm - mean (Std Err.), N: Nonclinical alarms/patient day	29.89(2.4), N = 71	26.34(6.38), N = 61	-	-	-
	Short duration nonclinical alarms/patient day	22.63(1.81), N = 71	14.32(1.9), N = 61	-	-	-
	Long duration nonclinical alarms/patient day	2.67(0.26), N = 71	1.58(0.2), N = 61	-	-	-
	Duration of nonclinical alarms/patient day	1679.53(185.69), N = 71	56084.88(15639.76), N = 61	-	-	-
	Nonclinical alarms/monitored hour	6.39(0.8), N = 71	1.0(0.4), N = 61	-	-	-
	Short duration nonclinical alarms/monitored hour	4.84(0.63), N = 71	0.54(0.15), N = 61	-	-	-
	Long duration nonclinical alarms/monitored hour	0.57(0.14), N = 71	0.06(0.02), N = 61	-	-	-
	Duration of nonclinical alarms/monitored hour	359.24(42.78), N = 71	2132.4(676.96), N = 61	-	-	-
New et al., 2016	NOTECHS Mean (SD) - a measure of operating teams' non-technical skills	77.84 (11.59), N = 25	78.06 (6.57), N = 16	-	-	*p=0.938*
	WHO Time Out attempt - Component of WHO surgical safety checklist – n/N (%)	24/25 (96%)	16/16 (100%)	-	-	*p=1.000*
	WHO Time-Out complete compliance - number of cases in which all three components of Time-Out were completed - n/N (%)	9/25 (36%)	10/16 (62%)	-	-	*p=0.621*

Supplementary file 2- Data extraction tables

	WHO Sign Out - number observed- n/N (%)	1/25 (4%)	1/16 (6%)	-	-	*p=1.000*
	Glitch rate/hour - these are deviations from recognised process with potential to reduce quality or speed – mean (SD), N	6.59 (SD 3.95), N = 25	7.94 (SD 4.01), N = 16	-	-	*p=0.098*
	Length of stay in days (SD), N	7.7 (SD 15), N = 292	7.6 (SD 16), N = 173	-	-	*p=0.396*
Rothemich et al.	Counselling behaviour:					
	Ask – patient was asked if you smoke	*525*/857 (61.2%)	*637*/958 (66.5%)	-	-	*p=0.45*
	Advise- patient was advised to stop	*499*/857 (58.2%)	*530*/958 (55.3%)	-	-	*p=0.39*
	In-office cessation support(unadjusted)	*349*/857 (40.7%)	*270*/958 (28.2%)	-	-	*p<0.001*
	In-office cessation support (adjusted)	*333*/857 (38.9%)	*273*/958 (28.5%)	-	-	
	Patient who had discussion	*295*/857 (34.4%)	*262*/958 (27.4%)	-	-	*p=0.001*
	Patients who had referral to quitline	*183*/857 (21.4%)	*83*/958 (8.7%)	-	-	*p<0.001*
Rustagi et al., 2016	Mean proportion of pregnant women tested for HIV at antenatal care visit:	Overall mean=95.9%, N = 18	Overall mean=93.4%, N=18	-	-	-
	Cote d'Ivoire	100%, N = 6	99.9%, N = 6	-	-	-
	Kenya	96.0%, N = 6	87.3%, N = 6	-	-	-
	Mozambique	91.7%, N = 6	92.9%, N = 6	-	-	-

Supplementary file 2- Data extraction tables

Quality assessment

Table 6: CASP questions for appraisal of a Cohort Study (possible answers for each question are: Yes, Can't Tell and No)

		Studies	1. Did the study address a clearly focused issue?	2. Was the cohort recruited in an acceptable way?	3. Was the exposure accurately measured to minimise bias?	4. Was the outcome accurately measured to minimise bias?	5. (a) Have the authors identified all important confounding factors?	5. (b) Have they taken account of the confounding factors in the design and /or analysis	6. (a) Was the follow up of subjects complete enough?	6. (b) Was the follow up of subjects long enough?	7. Do you believe the results?	8. Can the results be applied to the local population?	9. Do the results of this study fit with other available evidence?	10. What are the implications of this study for practice?
1	**Cohort**	**Afsar-manesh et al.**[1]	Y	Y	Y	Y	CT	Y	Y	Y	Y	Y	Y	Y
2	**Case Control**	**Allaudeen et al.**	See	Next										
3	**Cohort**	**Anderson et al.**	Y	Y	Y	Y	Y	CT	Y	Y	Y	Y	Y	Y
4	**Cohort**	**Bell et al.**	Y	Y	Y	Y	Y	CT	CT	N	Y	CT	Y	Y
5	**Case Control**	**Bhatt et al.**[3]	See	Next										
6	**Case Control**	**Bhutani et al.**[4]	See	Next										
7	**Cohort**	**Bowen et al.**	Y	Y	Y	Y	CT	CT	N	N	Y	CT	Y	Y
8	**Cohort**	**Bradley et al.**[5]	Y	Y	Y	Y	CT	CT	Y	Y	Y	N	Y	Y
9	**Case Control**	**Catchpole et al.**[6]	See	Next										
10	**Cohort**	**Chandrasekar et al.**	Y	Y	Y	Y	Y	CT	CT	Y	CT	CT	CT	CT
11	**Cohort**	**Cochran et al.**	Y	Y	Y	Y	N	N	CT	Y	Y	CT	CT	Y
12	**Cohort**	**DeFlitch et al.**	Y	Y	Y	Y	Y	CT	CT	Y	Y	CT	Y	Y
13	**Cohort**	**Dennerlein et al.**[7]	Y	Y	Y	Y	CT	CT	Y	Y	Y	N	Y	Y
14	**Cohort**	**Gupta et al.**	Y	Y	Y	Y	CT	CT	CT	Y	Y	Y	Y	Y
15	**Cohort**	**Hathout et al.**[8]	Y	Y	Y	CT	Y	CT	Y	CT	Y	Y	CT	Y
16	**Cohort**	**Heymann et al.**[9]	Y	N	CT	N	N	N	CT	CT	CT	CT	CT	N
17	**Cohort**	**Hultman et al.**	Y	Y	Y	Y	Y	CT	CT	Y	Y	Y	Y	Y
18	**Cohort**	**Huntington et al.**[10]	Y	Y	N	Y	Y	Y	Y	CT	CT	N	Y	Y
19	**Cohort**	**Hwang et al.**[11]	Y	Y	Y	Y	Y	CT	Y	Y	Y	Y	Y	Y
20	**Cohort**	**Kane et al.**	Y	Y	Y	Y	Y	CT	CT	CT	CT	CT	CT	CT
21	**Cohort**	**Khan et al.**	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
22	**Cohort**	**Kottke et al.**[12]	Y	Y	Y	Y	Y	Y	Y	CT	Y	Y	Y	Y
23	**Cohort**	**Lick et al.**[13]	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
24	**Cohort**	**Loh et al.**[14]	Y	Y	CT	Y	Y	Y	Y	Y	Y	N	Y	CT
25	**Case Control**	**McGrath et al.**	See	Next										
26	**Cohort**	**McKetta et al.**[15]	Y	Y	Y	Y	CT	CT	Y	CT	Y	Y	Y	Y
27	**Cohort**	**Moran et al.**	Y	Y	Y	Y	Y	CT	Y	Y	Y	Y	Y	Y
28	**Cohort**	**New et al.**[16]	CT	Y	Y	CT	N	N	Y	N	N	CT	CT	CT
29	**Cohort**	**Rateb et al.**[17]	Y	Y	Y	Y	CT	CT	CT	CT	Y	Y	Y	Y
30	**RCT**	**Rothemich et al.**[18]	See	Two	Next									
31	**RCT**	**Rustagi et al.**[19]	See	Two	Next									

Supplementary file 2- Data extraction tables

32	**Cohort**		**Ryan et al.**[20]	Y	Y	CT	CT	CT	CT	Y	Y	CT	CT	Y	Y
33	**Cohort**		**Shultz et al.**[21]	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
34	**Cohort**		**Srinivasan et al.**	Y	Y	Y	Y	Y	CT	CT	N	CT	CT	CT	CT
35	**Cohort**		**Tetuan et al.**[22]	Y	Y	CT	CT	Y	CT	Y	CT	CT	Y	Y	CT

Table 7: CASP questions for appraisal of a Case Control Study (possible answers for each question are: Yes, Can't Tell and No)

		Studies	1. Did the study address a clearly focused issue?	2. Did the authors use an appropriate method to answer their question?	3. Were the cases recruited in an acceptable way?	4. Were the controls selected in an acceptable way?	5. Was the exposure accurately measured to minimise bias?	6. (a) Aside from the experimental intervention were the groups treated equally?	6. (b) Have the authors taken account of the potential confounding factors in the design and/or in their analysis?	7. Can this be applied to the local population?		8. Do the results fir with other available evidence?
2	**Case Control**	**Allaudeen et al.**	Y	Y	Y	Y	Y	Y	Y	CT		CT
5	**Case Control**	**Bhatt et al.**[3]	Y	Y	Y	N	Y	Y	Y	Y		CT
6	**Case Control**	**Bhutani et al.**[4]	Y	Y	Y	Y	Y	Y	Y	Y		Y
9	**Case Control**	**Catchpole et al.**[6]	Y	Y	Y	Y	Y	Y	Y	Y		Y
25	**Case Control**	**McGrath et al.**	Y	Y	Y	CT	Y	CT	Y	Y		Y

Table 8: CASP questions for appraisal of RCTs (possible answers for each question are: Yes, Can't Tell and No)

		Studies	1. Did the trial address a clearly focused issue?	2. Was the assignment of patients to treatments randomised?	3. Were all the patients who entered the trial properly accounted for at its conclusion?	4. Were patients, health workers and study personnel blind to treatment?	5. Were the groups similar at the start of the trial?	6. Aside fro the experimental intervention, were the groups treated equally?	7. Can the results be applied to the local population, or in your context?	8. Were all clinically important outcomes considered?	9. Are the benefits worth the harms and risks?
30	**RCT**	**Rothemich et al.**[18]	Y	Y	Y	N	Y	Y	Y	Y	Y
31	**RCT**	**Rustagi et al.**[19]	Y	Y	Y	N	Y	Y	Y	Y	Y

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Supplemental material

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